NATIONAL AVIATION ACADEMY OF THE
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MANAGEMENT SYSTEM TO ENHANCE ACCESSIBILITY AND ENGAGEMENT IN NATIONAL AVIATION ACADEMY OF THE PHILIPPINES - VILLAMOR

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Chapter 1
Introduction

Student organizations serve as vital pillars of student development in higher education, fostering leadership, collaboration, and civic engagement beyond the classroom, while universities worldwide are increasingly adopting integrated information systems to strengthen governance, streamline administrative processes, and improve stakeholder engagement. Research shows that educational information systems enhance accessibility to resources, institutional coordination, and student engagement through centralized digital platforms [1], while studies on educational technology adoption emphasize that usability, accessibility, and user acceptance are critical to successful implementation in higher education [2]. Despite this global shift toward intelligent campus environments and digital management, the National Aviation Academy of the Philippines–Villamor (NAAP–Villamor) still manages student organization processes through largely manual, decentralized, and fragmented procedures under the supervision of the Office of Student Affairs (OSA), where reports and proposals are submitted physically, approvals are processed manually, and communication occurs through separate platforms, resulting in delays, inconsistent documentation, limited transparency, and added workload for both student officers and OSA personnel. This condition contrasts with national directives such as CHED Memorandum Order No. 09, s. 2020, which promotes

Smart Campus development [3], Republic Act No. 11032, which mandates efficient and electronic service delivery in government entities [4], and ARTA Memorandum Circular No. 2020-03A, which reinforces zero-contact digital transactions [5]. As a result, the absence of a centralized and integrated management system tailored for student organizations at NAAP–Villamor highlights a clear gap between national digital governance policies and their actual institutional implementation.

The continued absence of a secure, centralized, and policy-aligned digital system creates operational inefficiencies and limits student engagement. Manual processes hinder real-time monitoring, delay approval workflows, and reduce accessibility to organization-related services. Recent studies on information system success in higher education emphasize that system quality, information quality, and service quality significantly influence user satisfaction and overall system effectiveness [6]. Therefore, the development of an Integrated Management System Enhancing Student Organization Member Accessibility and Engagement at NAAP–Villamor is necessary to centralize reporting, automate workflows, improve accessibility, and ensure compliance with national digital governance policies [3][4][5].

To identify the institutional gaps, this study formulates specific research questions that guide the design, development, and evaluation of the proposed system. These questions focus on system architecture, accessibility, policy compliance, and performance effectiveness.

This study seeks to answer the following research questions:

- How can the management system streamline the submission, tracking, and archiving of organizational documents in compliance with RA 11032 and ARTA policies?
- How can the system improve student engagement through centralized communication, activity management, and access to services?
- What is the quality of the management system based on the ISO/IEC 25010 model in terms of functionality, usability, reliability, performance efficiency, and security?
- What is the level of usability and acceptability of the system among students, organization officers, and administrators based on User Acceptance Testing (UAT)?
- How can a clear and comprehensive user manual support users in properly using and navigating the system?
- What recommendations and feedback from users can be gathered to further improve and enhance the system?

Project Context

Higher education institutions worldwide are increasingly integrating digital technologies to enhance governance, administrative efficiency, and institutional transparency. Universities are adopting information systems to support data management, streamline administrative workflows, and improve coordination across academic and operational units. Studies on digitalization in higher education show that institutions implementing integrated digital systems experience improvements in service delivery, information accessibility, and organizational coordination. Digital Transformation Theory, these improvements occur when digital technologies are embedded within institutional processes and governance structures rather than functioning as isolated tools. As higher education environments become more complex, reliance on manual and fragmented administrative procedures becomes inefficient, making digital transformation an essential strategy for effective institutional management.

In the Philippine context, higher education institutions are encouraged to align with national initiatives on digital transformation and efficient public service delivery. Policies promoting streamlined transactions and improved administrative processes emphasize the importance of adopting digital systems within academic institutions. Despite these initiatives, many institutions still experience challenges in fully implementing integrated platforms, particularly in managing student-related administrative functions. This creates a gap between policy direction and actual institutional practice.

In specialized sectors such as aviation education, administrative precision and systematic governance are particularly critical because aviation institutions operate within structured and compliance-oriented environments. Accurate documentation, coordinated reporting, and institutional accountability are essential components of administrative operations. Information Systems Governance frameworks emphasize that organizational processes should be supported by coordinated information systems that enable monitoring, accountability, and structured decision-making. Consequently, governance functions including the management of student organizations and co-curricular activities require structured systems that ensure transparency and consistent institutional oversight.

At the National Aviation Academy of the Philippines – Villamor (NAAP–Villamor), student organizations operate as formally recognized institutional entities under the supervision of the Office of Student Affairs (OSA). These organizations contribute to leadership development, student engagement, and co-curricular participation within the academy. Organizational governance typically involves student officers, faculty advisers, and administrative oversight from OSA, forming a structured governance flow that regulates activities, documentation, and compliance monitoring. However, despite the presence of established governance policies, the current operational environment largely relies on manual procedures, fragmented documentation systems, and dispersed communication channels rather than a centralized digital platform capable of systematically managing organizational processes.

This administrative environment creates several operational gaps that affect governance efficiency and institutional monitoring. Processes such as document submission, event coordination, activity monitoring, and rental service management are frequently conducted through manual procedures or isolated digital tools. Also, administrative systems in higher education indicates that fragmented documentation processes often lead to delays in approvals, difficulty tracking records, and inconsistencies in organizational. Similarly, paper-based administrative workflows increase the risk of misplaced records, inefficient verification procedures, and limited transparency in institutional monitoring. These limitations reduce administrative visibility and hinder the ability of institutional units to effectively oversee student organizational activities.

Within the broader research landscape, many universities have implemented digital platforms designed to support student organization management, campus event coordination, and institutional activity monitoring. Integrated management systems allow centralized scheduling, digital document submission, and administrative tracking of organizational activities, improving operational coordination and information accessibility. However, most existing systems are designed for general university environments and focus primarily on event coordination rather than comprehensive governance workflows. As a result, the design and evaluation of integrated student organization management systems within specialized institutions—such as aviation academies—remain relatively underexplored in current research.

Addressing this gap requires a systems-oriented approach that aligns governance policies with digital infrastructure. Institutional Theory explains that organizations develop structured systems and procedures to maintain legitimacy, accountability, and operational stability. Within this framework, implementing an Integrated Student Organization Management System enables governance policies to be operationalized through structured digital workflows. By integrating people, processes, and technology, the proposed web-based system aims to improve administrative efficiency, enhance transparency, and strengthen institutional oversight of student organizations within NAAP–Villamor.

Purpose and Description

This research aims to design, develop, and evaluate an integrated web-based management system for student organizations at NAAP–Villamor. It is an applied systems study that focuses on improving how student organization processes are managed through a centralized digital platform. The research is expected to contribute knowledge on how a web-based system can support governance, accessibility, and administrative efficiency in student organization management.

Digital management of student organization processes is also examined in this research to determine how it can improve institutional operations that are still handled manually. The work adds to existing studies on student organization management systems by exploring their use in a specialized educational setting such as an aviation academy. Through this, context-based knowledge may be provided on the role of integrated systems in higher education administration.

The study aims to address the inefficiencies in the current student organization procedures at NAAP–Villamor, which are largely dependent on manual and fragmented processes. These existing practices affect the timeliness, consistency, and accessibility of organizational records and administrative workflows. In response, the study proposes the development of a web-based integrated system designed to centralize information, streamline processes, and enhance access to organizational services. Through this approach, the system is intended to support more efficient coordination and improved administrative management within the Office of Student Affairs.

Guidance for the research is drawn from the Input–Process–Output model, which shows how the system is developed and how results are produced. For software quality measurement, the ISO/IEC 25010 model serves as the main framework for assessing the system.

The system covers core functions such as document management, submission and monitoring of reports, activity management, announcements, analytics, and access to organization-related services and information. Its main users are students, student organization officers, and Office of Student Affairs administrators, each using role-based access. The system is limited to recognized student organizations within NAAP–Villamor and does not include mobile deployment, in-system payment, real-time chat, or full integration with the official enrollment database.

The study offers practical contribution by providing a system that can improve record handling, workflow efficiency, and access to services. Its theoretical contribution lies in showing how integrated digital systems can support governance and system quality in student organization management. Its institutional contribution is its support for better compliance, stronger oversight, and improved administrative processes within NAAP–Villamor.

The effectiveness of the system will be measured through system evaluation and user-based assessment. The evaluators will include students, student organization officers, and Office of Student Affairs administrators. The main instruments for evaluation are ISO/IEC 25010 criteria and User Acceptance Testing to measure functionality, usability, reliability, efficiency, security, and user acceptability.

The study expects the system to improve accessibility, records consolidation, workflow speed, transparency, and administrative monitoring. In the long term, it may help the institution sustain a more organized, efficient, and policy-aligned way of managing student organizations. Overall, the study seeks to show that an integrated web-based platform can support better governance and more effective student organization management at NAAP–Villamor.

Contextual Framework

The Input–Process–Output (IPO) Model is a conceptual framework commonly used to explain how a system transforms resources into desired results. It presents a logical flow in which inputs are gathered, processed through a set of activities or procedures, and converted into outputs that address a specific need or problem. In research and system development, the IPO model is useful because it provides a clear and organized way of showing how a proposed system operates, what elements are needed for its implementation, and what results are expected from it. Because of its simple but structured format, the model is often applied in studies involving information systems, organizational processes, and technology-based solutions.

The input phase refers to all the resources, information, requirements, and foundational elements needed for the system or study to function properly. These may include data, user requirements, policies, standards, hardware, software, and other supporting materials that serve as the basis of development and operation. Inputs are important because they define what the system needs in order to perform effectively and respond to the identified problem or objective.

The process phase represents the set of activities, methods, and procedures used to transform the inputs into meaningful results. This stage includes the actual operations performed within the system, such as planning, designing, developing, organizing, analyzing, validating, or managing information.

In the context of information systems, the process phase is where inputs are handled systematically so that they can produce accurate, efficient, and useful outcomes.

The output phase refers to the final products, results, or outcomes generated after the inputs have been processed. These outputs may take the form of reports, services, improved workflows, decisions, or fully developed systems depending on the nature of the study. The output phase is significant because it reflects whether the system or process has successfully achieved its intended goals and delivered value to its users or stakeholders.

The feedback component is an important part that strengthens the Input–Process–Output (IPO) Model because it introduces continuous evaluation and improvement into the framework. While the basic IPO model shows how inputs are transformed through processes into outputs, feedback allows the results of the system to be reviewed and used as a basis for refinement. It helps determine whether the outputs meet the intended objectives, satisfy user needs, and comply with required standards or policies.

The Input–Process–Output (IPO) Model is best suited for this study because it clearly shows how the proposed system transforms essential inputs, such as user data, organizational records, policies, and quality standards, into meaningful outputs through structured system processes like design, authentication, monitoring, and evaluation. It is appropriate for the study because it provides a simple yet strong framework for explaining how the web-based management system is developed and how it improves accessibility, efficiency, transparency, and governance in student organization management.

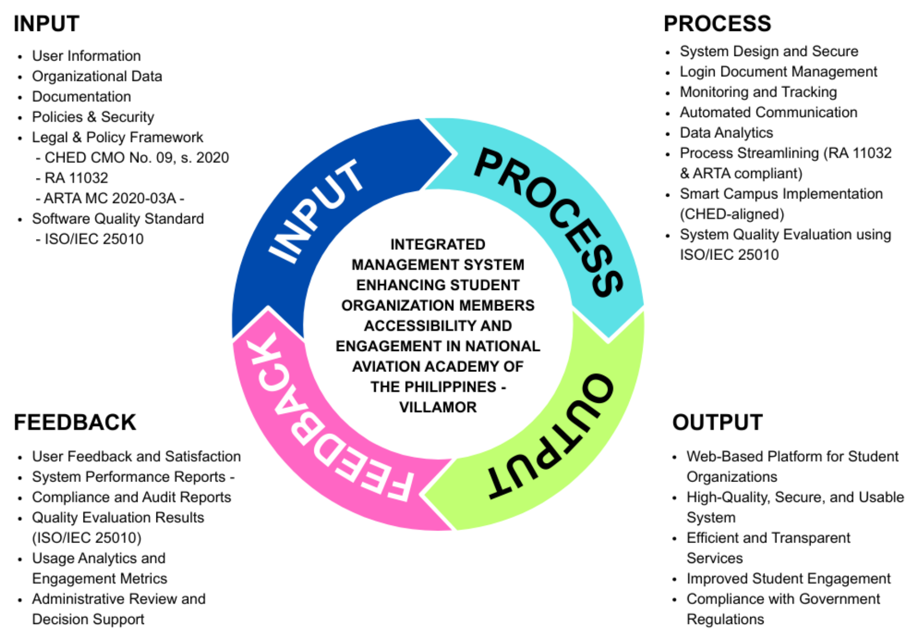

Figure 1. **Contextual Framework of the Study**

This conceptual framework follows the Input–Process–Output (IPO) Model, demonstrating how a web-based student organization management system improves coordination and governance among student organizations and the Office of Student Affairs. Key inputs such as user information, organizational data, institutional policies, and security requirements are processed through system design, user authentication, monitoring, data analytics, document approval, and notification services. These processes result in an integrated management system that provides organized records, analytics dashboards, and improved administrative oversight, ensuring an efficient, secure, and data-driven approach to student organization management.

The input phase represents the foundational elements required for the development of the proposed web-based management system. These inputs include user information, organizational data, and official reports which serve as the primary data sources necessary for managing student organization activities, membership records, documentation, and administrative transactions within the platform.

In addition to data resources, this phase incorporates legal and policy frameworks that guide the system's design and implementation. These include CHED Memorandum Order No. 09, Series of 2020, which promotes smart campus initiatives and the integration of information and communication technologies; Republic Act No. 11032 (Ease of Doing Business and Efficient Government Service Delivery Act), which mandates the Zero-Contact Policy and streamlined transactions; and ARTA Memorandum Circular No. 2020-03A, which requires the use of electronic platforms to reduce red tape and physical interactions. These policies ensure that the system aligns with national directives on digital transformation, efficiency, and transparency.

Furthermore, ISO/IEC 25010 is included as a quality standard input to guide system planning and design. This international standard defines quality characteristics such as usability, security, performance efficiency, reliability, and maintainability, ensuring that the system is not only compliant with policy requirements but also meets accepted software quality benchmarks.

The process phase covers the transformation of inputs into a fully functional system through a structured and systematic approach. This includes system design, development, implementation, testing, and evaluation. During this phase, key system features such as secure login mechanisms, digital document management, activity monitoring, automated communication, and data analytics are designed and integrated.

The processes are explicitly aligned with RA 11032 and ARTA MC 2020-03A by automating transactions, minimizing manual interventions, and reducing the need for physical office visits, thereby operationalizing the Zero-Contact Policy. Approval workflows, document submissions, and monitoring activities are streamlined to improve efficiency and transparency while reducing administrative bottlenecks. At the same time, the system supports CHED’s Smart Campus initiative by enhancing accessibility to student services and organizational information through a centralized digital platform. To ensure software quality, ISO/IEC 25010 is applied during testing and evaluation, covering functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. This ensures that the system performs effectively, securely, and consistently across different user roles and environments.

The output phase represents the resulting product of the study: a centralized, web-based management system for student organizations at NAAP–Villamor. The system provides secure digital record keeping, efficient monitoring of organizational activities, and automated administrative services that support faster and more transparent processes.

Through features such as analytics dashboards and centralized information access, the platform enables data-driven decision-making, improves coordination between student organizations and the Office of Student Affairs, and enhances overall student engagement. The output aligns with CHED policies, complies with ease-of-doing-business and anti-red tape regulations, and meets ISO/IEC 25010 standards for usability, reliability, and security, ensuring that the system is both functional and institutionally compliant.

The feedback mechanism ensures continuous system improvement and long-term sustainability. Feedback is collected through user satisfaction surveys, system performance reports, usage analytics, compliance and audit reports, and ISO/IEC 25010 evaluation results. These feedback inputs are analyzed to identify areas for enhancement, improve system efficiency and transparency, strengthen security and usability, and ensure ongoing compliance with legal, policy, and quality standards. This continuous feedback loop allows the system to adapt to evolving institutional needs and policy requirements, thereby maintaining its effectiveness as a digital platform that supports Zero-Contact, efficient service delivery, and good governance in student organization management.

Objectives of the Study

This study aims to design, develop, and evaluate a Management System that enhances accessibility and engagement of student organizations at NAAP–Villamor. Specifically, the study aims to:

1. To design and implement system features that streamline the submission, tracking, and archiving of organizational documents in compliance with RA 11032 and ARTA policies.
2. To develop system functionalities that improve student engagement through centralized communication, activity management, and access to services.
3. To evaluate the system's quality using the ISO/IEC 25010 model in terms of functionality, usability, reliability, performance efficiency, and security.
4. To assess the system's usability and acceptability through User Acceptance Testing (UAT) among students, organization officers, and administrators.

5. To create a clear and comprehensive user manual that will guide users on how to properly use and navigate the system.

6. To gather valuable recommendations and feedback from users in order to improve and enhance the system.

Scope and Limitation

This study covers the design, development, and evaluation of an integrated web-based management system intended to enhance the accessibility, coordination, and management of student organization-related processes at the National Aviation Academy of the Philippines–Villamor Campus. It focuses on student organizations formally recognized by the Office of Student Affairs and includes currently enrolled students, student organization officers, and OSA administrators as the primary users of the proposed system. The study is conducted within the institutional setting of NAAP–Villamor and is limited to the organizational processes, needs, and structure existing within the academy.

Specifically, the study covers the digital management of organization-related records, submission and monitoring of documents, organization activity management, dissemination of announcements, and access to organization-related services and information through a centralized web-based platform. It also includes analytics features intended to support the interpretation, monitoring, and assessment of organization-related data, including AI-assisted analytics where applicable. In addition, the study covers the evaluation of the proposed system in terms of functionality, usability, and user acceptability as perceived by its intended users within the institution.

The proposed system is limited to a web-based implementation and requires internet connectivity for access and operation. It does not directly integrate with the institution's official enrollment database; therefore, real-time verification of students' current enrollment status is not implemented and is assumed to be handled externally by the institution. Although the system includes analytics and AI-assisted analytics as part of its functionality, the personalization of notifications and analytics remains basic and depends largely on user data input. Communication with administrators for technical support or special requests is also handled outside the system, typically through email or other messaging platforms. While the system works on major web browsers, performance or display may vary slightly depending on browser type and version. The effectiveness of the system may also depend on the availability of technological resources, the stability of internet access, and the accuracy and completeness of data entered by authorized users.

The system focuses on currently enrolled students, student organization officers, and administrative personnel from the Office of Student Affairs, which may limit its relevance to other members of the school community or external users. The initial deployment may involve only a limited number of organizations and users for testing purposes, which may restrict the diversity of services and events available at the start. In its initial version, the system does not cover mobile application deployment, in-system payment processing, real-time chat features, or other advanced third-party integrations.

The platform is limited to the National Aviation Academy of the Philippines–Villamor Campus. As such, students or organizations outside the institution cannot participate in or access organizational services through the system. Thus, the findings of the study are limited to the implementation and evaluation of the system within the specified institutional context.

Chapter 2

REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and studies that provide the theoretical and empirical foundation of the present research. The selected sources discuss concepts and systems relevant to the proposed integrated management system, particularly in the areas of artificial intelligence, document and records management, authentication, user management, and role-based access control. These topics are essential in establishing the relevance of digital solutions in supporting institutional processes and service delivery.

The reviewed literature also shows how technology-based systems contribute to improving accessibility, efficiency, security, and accuracy in managing organizational and administrative functions. Studies on notification systems, centralized data handling, and secure access mechanisms demonstrate the value of digital platforms in streamlining workflows, strengthening monitoring, and protecting user information. These insights are significant in understanding the functions and features that may support the needs of student organizations and their members.

To support the development of the proposed system for the National Aviation Academy of the Philippines – Villamor, this chapter examines related works that highlight the role of integrated platforms in enhancing student engagement, administrative responsiveness, and service accessibility. The gathered literature serves as a guide in identifying relevant technologies, system practices, and operational approaches that may contribute to the design and implementation of the study.

Artificial Intelligence in Data Analytics

Artificial intelligence in data analytics has become increasingly important in educational settings because it enables institutions to process large amounts of data and transform them into useful information for monitoring, prediction, and decision-making. In higher education, AI-driven analytics can help identify trends, interpret user behavior, and improve institutional processes and student-related services. This is relevant to the present study because the proposed system for the National Aviation Academy of the Philippines – Villamor includes centralized records, role-based monitoring, digital workflows, and analytics dashboards intended to improve accessibility, oversight, and student engagement.

One key application of artificial intelligence in data analytics is the use of dashboards that provide actionable insights rather than simply presenting raw data. Learning analytics dashboards are most effective when they help users interpret information and support decision-making [7]. This is relevant to the present study because the proposed system includes analytics dashboards and reporting features that may help administrators and student organization officers monitor activities, records, and service utilization more effectively.

Artificial intelligence has also been shown to improve engagement and performance when combined with predictive analytics. Integrating AI performance prediction with learning analytics enhanced student engagement, collaborative learning, and satisfaction [8]. Learning analytics can also be used to examine student engagement and improve performance [9]. These findings support the idea that the proposed system may be used not only for record management but also for monitoring participation and identifying areas that may require intervention.

The relevance of AI-driven analytics is further supported by studies on digital management systems. Artificial intelligence in learning management systems contributes to more adaptive and responsive digital environments [10]. In the same way, AI and educational data mining can identify patterns and user needs, making systems more evidence-based and responsive [11]. These findings are closely related to the present research because the proposed system also seeks to centralize information and improve workflows through data-informed processes.

Recent studies likewise show the growing role of machine learning and generative AI in higher education analytics. These technologies are increasingly used for personalized feedback, trend analysis, and early risk detection [12]. This is especially relevant to the present capstone project because the proposed integrated management system is intended not only to store information but also to support monitoring, reporting, and data-driven decision-making.

Overall, the reviewed studies show that artificial intelligence in data analytics can strengthen digital management systems by transforming raw data into actionable information for monitoring, prediction, and user support. These findings support the present capstone project by showing that analytics dashboards, predictive tools, and AI-supported analysis can improve engagement, strengthen oversight, and enhance system responsiveness. Such insights are aligned with the objectives of the proposed system for the National Aviation Academy of the Philippines – Villamor, particularly in improving accessibility, streamlining workflows, and supporting data-driven governance.

Document and Record Management

Document and Records Management (DRM) in higher education has increasingly been treated as an institutional governance function rather than a clerical task, especially as universities and colleges digitize administrative workflows and accountability requirements. Literature from 2022 to 2026 shows that paper-heavy and fragmented storage practices weaken monitoring, slow down approvals, and make retrieval unreliable—issues that become more evident in offices handling high volumes of compliance documents such as endorsements, approvals, activity reports, and financial records.

Digital transformation in records management improves efficiency and accuracy, particularly when records are consolidated into a centralized process rather than scattered across manual systems [13]. Centralization increases visibility, strengthens coordination, and reduces time spent locating documents. A campus-level document management system enhances organization and accessibility by improving how records are captured and stored, enabling faster retrieval for administrative action [14]. Manual or partially digitized recordkeeping leads to tracking inconsistencies and redundancy, while structured e-document systems reduce these issues through organized storage and retrieval [15]. These studies show that centralization is not only a technical improvement but also strengthens institutional memory and operational continuity.

Beyond system development, studies emphasize that modern DRM requires structured models, policies, and best practices to sustain digitization. Electronic document management benefits from established frameworks, including policy guidance and digital preservation, indicating that simple folder-based storage is insufficient for reliable governance [16]. Audits of web-based document and records systems highlight the need for oversight, as systems must support archiving, retrieval, and reviewability [17]. This reinforces that DRM solutions must integrate monitoring and quality assurance mechanisms to maintain institutional integrity.

However, literature also notes that digitization alone does not resolve records management issues. Barriers such as gaps in strategy, limited skills, and governance challenges can hinder successful implementation [18]. Even with available technologies, effectiveness depends on leadership support, training, infrastructure, and sustained adoption [19]. These findings emphasize that centralization works best when supported by institutional readiness and alignment.

Security and compliance remain essential in DRM systems, especially when handling sensitive information. Organizations must ensure confidentiality, integrity, and availability through proper security controls, supporting access restrictions and traceability [20]. Education information systems serve as institutional infrastructure for coordination and monitoring, reinforcing that DRM must support governance and reliability, not just convenience [21].

Overall, literature from 2022 to 2026 highlights three key points: centralized repositories improve retrieval and monitoring; governance models and audits ensure sustainability and oversight; and successful digital transformation depends on readiness, capacity, and security. These insights support the present study's focus on improving document-based monitoring within the Office of Student Affairs. Effective DRM requires centralized capture, organized storage, fast retrieval, and safeguards to prevent loss or unauthorized access, ensuring accountability through structured and secure recordkeeping.

Indexing and Query Optimization

Indexing and query optimization are database techniques aimed at improving the efficiency of data retrieval operations. B-Tree indexing provides a hierarchical structure for storing data, allowing for rapid searches, inserts, and deletions even in large datasets [22]. In the context of a student organization management system, indexing could allow quick retrieval of attendance logs, membership records, or document submissions without causing delays during peak usage.

Research supports the application of indexing for large-scale administrative systems. B-Tree indexing is effective in accelerating queries for frequently accessed columns, such as user IDs, booking IDs, and document codes [22]. Cursor-based pagination allows users to navigate through large datasets efficiently, maintaining system performance even when handling extensive logs or reports [23]. These findings suggest that indexing and optimized query strategies are recommended for high-volume data environments like centralized student service platforms.

In relation to this study, indexing and query optimization could facilitate fast access to attendance records, document histories, and event participation data. Efficient data retrieval enhances user experience for students, officers, and administrators, allowing seamless filtering, searching, and reporting within the system [22][23]. Applying these techniques ensures that the system remains responsive and scalable, even when handling multiple submissions, high attendance logs, or simultaneous user access across the student organization platform.

Notification Systems

Notification systems deliver announcements, reminders, and alerts to specific users through channels such as in-app messages, dashboards, email, SMS, or push notifications. In campus settings, they help reduce delays, consolidate fragmented communication, and provide traceable records of what was sent and when [24][25]. This section explains how notifications address common communication issues in student organization environments and how prior studies support a centralized notification feature that improves awareness, follow-ups, and accountability among students, organization officers, and administrative offices. However, existing studies primarily focus on general campus-wide communication and do not fully address notification systems integrated within student organization governance workflows, particularly in ensuring real-time coordination, role-based targeting, and process-specific alerts in specialized institutional settings.

Studies emphasize that timely dissemination is critical for coordination and participation, especially when announcements are scattered across multiple platforms [24]. Delivery design—such as audience targeting, timing, and relevance—strongly influences engagement and whether users treat notifications as useful updates or as noise [25]. Together, these findings support a centralized announcement function that targets specific user groups, reducing delays, repeated reposting, and manual follow-ups [24][25].

Centralized noticeboard portals further improve reliability by providing a single authoritative source of announcements while enabling faster delivery through integrated messaging options [25][26]. However, several implementations remain standalone or limited to one delivery channel, which weakens coverage and continuity. Linking notifications with automated attendance or smart check-in mechanisms strengthens the connection between announcements and actual participation, though such approaches may require added infrastructure and compliance [27]. These studies point to an integrated platform where announcements, events, and participation tools work together so offices can promote activities and monitor engagement within one system [25][27].

For event awareness, reminders are most effective when they are scheduled and targeted, using mechanisms such as countdown alerts and triggered notifications. Evidence from event management implementations suggests that combining web interfaces with SMS or push reminders can increase attendance and user satisfaction, especially when RSVP actions are included [28][29]. User-centered timing and relevance reduce perceived spam and support sustained engagement [25]. These findings support reminders that respond to user actions, such as RSVP confirmation and follow-up after no response, rather than relying only on generic schedules [25][27][28].

Notifications are also widely used for workflow transparency through submission and approval status alerts, such as submitted, under review, approved, and rejected. These reduce repeated manual follow-ups by giving users automatic updates whenever a record changes state. Studies on online noticeboards and workflow systems indicate that milestone alerts can reduce turnaround time and status inquiries when notifications are triggered by approval progress [26][30]. This supports automated, context-rich alerts that point users to the exact record and clarify required next steps, improving accountability for approvers and clarity for submitters [26][30].

Overall, the literature indicates that notification systems are most effective when centralized, role-guided, and embedded into the actual processes users perform—posting announcements, scheduling events, tracking participation, and monitoring approvals [24][26][28]. While studies consistently report gains in timeliness and communication efficiency, recurring limitations include single-channel delivery and standalone implementations that are not linked to records and workflow data. Building on these findings, this study supports an integrated notification approach that strengthens communication between students, organization officers, and administrators, reduces manual follow-ups, improves awareness, and promotes more accountable and efficient service delivery.

Role Based Access

Role-Based Access Control (RBAC) assigns system permissions according to user roles rather than individual users. This ensures users can only access the functions and information required by their responsibilities, strengthening security, accountability, and operational efficiency [31]. RBAC is especially important in higher education platforms where multiple stakeholders—students, student organization officers, and administrative offices—operate within one system. Without clear role boundaries, systems become harder to manage, more vulnerable to unauthorized actions, and less reliable for monitoring and service delivery.

Research shows that weak or inconsistent authorization leads to both security and operational problems. Inconsistent authorization practices are linked to fragmented workflows and increased risk of unauthorized access, which is relevant to systems involving submissions, reviews, and approvals [31]. Unclear role definitions often create over-privileged users, making it difficult to restrict functions and trace actions [32]. Although institutional contexts differ, these studies consistently show that poor access structures reduce traceability and weaken accountability when multiple roles share connected but distinct tasks.

To address uncontrolled access, studies identify RBAC as a practical solution for limiting overreach and improving system control. Integrating RBAC into web-based systems restricts actions to role-appropriate functions, reducing security risks and improving manageability, though effectiveness depends on well-designed roles and strict enforcement [32]. RBAC has also been demonstrated in enterprise platforms where administrators can monitor activity and manage applications without exposing sensitive controls to unauthorized users, though enterprise environments may not always match the resources and governance maturity of academic institutions [33]. Evidence from other domains reinforces RBAC's value for confidentiality: hierarchical RBAC protects sensitive information and supports system reliability [34], while separating student and administrative access improves efficiency in school management, although the undated nature of one study raises uncertainty about its recency [35].

Beyond basic restriction, RBAC strengthens governance by separating responsibilities across workflows that involve submission, review, approval, and oversight. When one role can perform conflicting actions, errors and bypassing controls become more likely, weakening accountability. Separation-of-duties constraints have been addressed through blockchain-based RBAC models, though such designs may introduce complexity for institutions prioritizing practical deployment [36]. In a context directly aligned with this study, hierarchical RBAC in a student organization document management system distinguishes roles for submission, review, approval, and oversight, with reported improvements in efficiency, transparency, and traceability, though results may depend on implementation scope and user compliance [37].

Local studies further support RBAC in institutional systems managing sensitive records and transactions. Role-based portals in a scholarship management system restrict read and write operations by responsibility and report strong evaluation results for security, reliability, and usability, even if the role set must be adapted for student organization workflows [38]. The absence of role-based permissions has also been identified as a privacy and security limitation in an applicant tracking system, with RBAC recommended for integration even though post-implementation outcomes were not reported [39].

RBAC also enables administrative oversight while protecting records from unnecessary exposure. RBAC-enabled dashboards support monitoring across subsystems while maintaining strict access boundaries [33]. RBAC has likewise been proposed for an accreditation databank system to address vulnerabilities in legacy document storage and strengthen record integrity, though proposal-based studies may provide limited real-world performance evidence [40]. Still, these works underline that when many users access institutional records, role-based restrictions are essential to protect privacy and integrity.

Overall, the literature positions RBAC as a foundational mechanism for multi-user institutional systems because it limits unauthorized actions, clarifies responsibility boundaries, supports separation of duties, and enables secure oversight [31][40]. While many studies are domain-specific, the consistent implication is that institutions must design RBAC around their actual workflow roles and enforce it through clear policies. Applied to student organization governance, RBAC can support controlled submissions and approvals, traceable monitoring, and accountable transactions—improving efficiency and transparency while protecting sensitive institutional records.

Rule-Based Workflow Logic

Rule-based workflow logic refers to a type of automation in information systems where predefined if–then rules dictate system behavior in response to specific conditions. Rule-based systems allow consistent decision-making, as outcomes are determined by evaluating explicit rules rather than relying on human judgment [41]. In the context of student organization management, such logic can help ensure that document approvals, report submissions, and membership requests follow a consistent procedure. For example, the system can automatically reject duplicate submissions or track pending approvals, aligning processes with institutional policies.

Studies show the effectiveness of rule-based workflow logic in administrative systems. Rule-based systems are particularly effective in environments that require compliance and auditability, where structured and predictable outcomes are essential [42]. Rule-based automation also reduces manual effort and human errors in repetitive administrative tasks, increasing operational efficiency [43]. These findings suggest that such logic could be recommended for systems designed to centralize document approvals, notifications, and activity tracking.

Rule-based workflow logic could support automated handling of student organization submissions, event approvals, and membership requests. Using predefined rules ensures consistent treatment of all submissions and aligns system responses with institutional regulations, such as RA 11032 and CHED's Smart Campus guidelines [41][43]. The logic can be leveraged to streamline processes, prevent duplicate entries, and provide automated alerts to officers or administrators, enhancing workflow efficiency and compliance within a centralized platform.

Student Service Utilization

Student service utilization is an important concern in higher education because the presence of institutional services does not automatically guarantee that students will access or benefit from them. The extent to which these services are utilized is influenced by several factors, such as accessibility, awareness, convenience, service quality, user satisfaction, and the efficiency of delivery systems. In the context of the present study, this concept is highly relevant because the proposed web-based student organization management system is intended to provide a centralized platform for organization-related services, records, requests, and information. Through this platform, the study aims to improve student access, simplify service processes, and strengthen administrative coordination within the National Aviation Academy of the Philippines – Villamor.

Studies have shown that the use of student support services has a meaningful relationship with student success and persistence in higher education. Campus service utilization can positively influence student outcomes, although patterns of use may vary depending on the needs and circumstances of learners [44]. This suggests that services become more effective when students are able to access them conveniently and efficiently. In relation to the present study, such findings support the development of a centralized and organized digital platform that may increase the utilization of student organization services, encourage greater participation, and reduce barriers associated with fragmented and manual processes.

The quality and structure of support services also play a significant role in shaping students' satisfaction with institutional systems. In a study on online learning environments, cognitive, emotional, and management support services positively influenced students' learning satisfaction [45]. This is relevant to the current study because it demonstrates that properly organized and well-delivered services can improve how students perceive and engage with institutional platforms. A more accessible and structured digital environment for student organization services may therefore result in greater convenience and satisfaction among users.

In addition, the quality of electronic services has been identified as a crucial factor affecting students' willingness to use institutional platforms. Elements such as website efficiency, the quality of information provided, and confidentiality of user data significantly contribute to user satisfaction and positive perceptions of digital service delivery. These dimensions of e-service quality are essential in higher education settings where students increasingly rely on online systems [46]. This is highly applicable to the proposed system, which seeks to centralize access to documents, requests, services, and monitoring tools. For this reason, maintaining usability, reliability, and security is essential in fostering student trust and encouraging continued system use.

Student satisfaction with digital systems has likewise been found to depend greatly on ease of use, system performance, and the quality of displayed information. These factors significantly influence whether users are satisfied with a system and whether they intend to continue using it over time [47]. This insight is particularly important to the present capstone project because the success of the proposed integrated management system will depend not only on initial adoption but also on the sustained use of the platform by students and organization officers. Therefore, ensuring that the system is user-friendly, responsive, and informative is necessary for promoting long-term student service utilization.

The improvement of student services has also been associated with broader institutional outcomes such as student satisfaction, retention, well-being, and overall success. Improving student services contributes to a more supportive and effective educational environment [48]. This reinforces the relevance of the present study, as the proposed system also aims to improve the delivery of student organization services through a more systematic and efficient digital platform. By making such services more visible, accessible, and manageable, the system may contribute to stronger student engagement and a more positive institutional experience.

Evidence from the Philippine higher education setting further strengthens the relevance of digital service systems in improving utilization. Students are more likely to use a platform when it is perceived as useful, easy to use, and responsive to their needs. Perceived usefulness strongly influenced satisfaction and intention to use, while ease of use contributed to the acceptance of the system as a practical and beneficial tool [49]. This is especially relevant to the current research because the proposed student organization management system is likewise expected to improve student service utilization by making access to organization-related services, records, requests, and updates more efficient and convenient for students.

Token-Based Authentication (JWT + HMAC-SHA256)

Token-based authentication using JSON Web Tokens (JWT) and HMAC-SHA256 is a method for secure and efficient identity verification in distributed systems. JWT allows self-contained tokens to carry all necessary user claims, which reduces the need for repeated database lookups during authentication [50].

In a student organization management system, token-based authentication can provide secure login for students, officers, and administrators while supporting role-based access to dashboards, documents, and reports.

Research supports the effectiveness of JWT with HMAC-SHA256 in secure systems. This approach ensures both integrity and authenticity of user tokens, preventing tampering and unauthorized access [50]. JWT also allows stateless authentication, which is particularly valuable in systems with multiple roles and frequent user interactions [51]. Studies indicate that JWT-based authentication is widely recommended for systems requiring secure, multi-role access to sensitive information.

In this study, token-based authentication could enhance the security and privacy of student and organization data while enabling role-based access across the platform. Implementing this algorithm can ensure that students, officers, and administrators have appropriate permissions to view, submit, or approve records, while unauthorized access is prevented [50][51]. This aligns with national guidelines such as RA 11032 and ARTA MC 2020-03A, where secure, digital, and zero-contact processes are emphasized, thereby supporting trust, compliance, and accessibility within the system.

User Management

User Management in academic systems has become more than handling usernames and passwords; it now functions as a governance mechanism that protects institutional systems by controlling who can access the platform, what actions each user can perform, and how user actions can be traced for transparency. In environments such as student organization monitoring systems, multiple stakeholders interact—students, officers, moderators, and administrators—making access control, identity reliability, eligibility validation, and auditability essential.

Digital identity systems should be designed around identity assurance, authentication strength, and risk-based controls that reduce unauthorized access and credential compromise [52]. This strengthens the argument that higher education platforms must move beyond weak authentication and incorporate stronger identity practices, especially when systems store sensitive information or become tools for compliance monitoring. Information security management also requires access control and protection of confidentiality, integrity, and availability [20], reinforcing that user governance must be embedded within system design rather than handled informally.

Recent research reinforces these standards by showing how institutional access systems need both usability and security. Single sign-on adoption in academic communities raises important security awareness issues, indicating that institutions must pair centralized login systems with user education, privacy considerations, and governance rules to reduce risky behaviors that can undermine security [53]. Integrating single sign-on can improve login efficiency and user satisfaction, supporting the practical argument that user management must balance security controls with user workflow needs to reduce friction and prevent workarounds [54]. These studies agree that centralized identity systems can reduce “credential sprawl,” but they also show that security outcomes depend on user behavior and policy enforcement, not only technology.

Role-based authorization and privilege control appear consistently as a core element of defensible user governance. Role-based access control in educational systems, especially when combined with stronger identity mechanisms, helps ensure that only legitimate, enrolled users access protected resources, supporting least-privilege principles in academic platforms [55]. Role-based access control models also formalize authorization by mapping permissions to roles, minimizing excessive privileges and reducing access-related errors [31]. The literature converges on the view that role assignment should be systematic rather than discretionary because ad hoc privilege granting increases the risk of misuse and operational confusion.

The literature also emphasizes traceability and monitoring as necessary complements to authentication and role assignment. Audit trails and anomaly detection strengthen oversight by enabling organizations to detect suspicious activities and policy violations [56]. This supports the governance argument that user management must make actions reviewable, especially in contexts where disputes, document approvals, or compliance checks occur. Decentralized identity models can enhance privacy and user control, though integration challenges remain for institutions [57]. This contrasts with centralized IAM designs that emphasize institutional control and consistent policy enforcement, showing a key tension between privacy-centered identity ownership and administration-centered governance.

The reviewed sources collectively support an integrated user management approach for student organization platforms: stronger authentication to reduce unauthorized access; role-based permissions to ensure proper responsibility boundaries; and auditing to ensure accountability. The studies also reveal important limitations: security gains depend on policy enforcement and user awareness, and identity architectures must be designed to fit institutional readiness and governance expectations. These findings connect directly to the present study because student organization monitoring involves sensitive records, approvals, and participation logs that require controlled access, role clarity, eligibility checks, and traceable activity logs to support transparency and institutional accountability.

Synthesis

The reviewed literature from 2022 to 2026 shows that digital management systems in higher education are important for improving governance efficiency, accessibility, and institutional accountability. Across areas such as analytics, records management, notification systems, authentication, workflow automation, and user management, the studies consistently show that fragmented and manual processes reduce transparency, delay services, and weaken monitoring. Overall, the literature points to the need for integrated platforms that centralize records, automate workflows, and support data-driven decision-making.

Several common findings emerge from the literature. First, centralization improves accessibility, retrieval, coordination, and monitoring. Second, governance and controlled access are essential, as role-based permissions and authentication help protect confidentiality, define responsibilities, and support traceability. Third, automation and analytics improve oversight by turning raw data into useful insights and standardizing processes such as approvals and monitoring. Finally, performance and usability influence continued system use, showing that both technical efficiency and user experience are necessary for successful digital platforms.

Despite these shared conclusions, the literature also reveals limitations. Many studies focus on individual components rather than fully integrated systems, and some do not directly connect analytics with governance workflows. In addition, implementation readiness remains a challenge, since organizational capacity, training, governance maturity, and policy enforcement affect whether digital systems succeed in practice. These gaps show that many institutional platforms still treat records management, authentication, workflow automation, and analytics as separate subsystems rather than as one unified governance tool.

The present study responds to this gap by proposing an Integrated Student Organization Management System for the National Aviation Academy of the Philippines – Villamor. The proposed system combines document management, authentication, notifications, analytics, and workflow logic into one centralized web-based platform. By doing so, it aims to improve governance efficiency, regulatory compliance, administrative effectiveness, and student service accessibility under a single supervisory framework. Overall, the study builds on the literature by promoting a centralized, secure, and analytics-supported platform for student organization management.

Chapter 3

RESEARCH AND DEVELOPMENT METHODOLOGY

This study utilized the Research and Development (R&D) methodology to guide the systematic creation and evaluation of the proposed system. The R&D approach is appropriate for this research because it focuses on developing a functional software product while ensuring that it effectively addresses the identified institutional problems. Through this methodology, the researchers were able to transform the gathered requirements into a structured system design and working application. It also allowed continuous refinement of system components through iterative testing and validation. This process ensured that both user requirements and institutional standards were consistently incorporated throughout development.

The study integrates research-driven analysis and software engineering principles to guide the creation of a scalable, secure, and efficient web-based system. It describes the selected Software Development Life Cycle (SDLC) model, development phases, system validation procedures, feasibility analysis, system architecture design, database modeling, security mechanisms, and implementation environment. Through this structured framework, the methodology ensures that the proposed system meets functional, technical, operational, and quality standards.

Software Development Life Cycle Model

This study adopted the Rapid Application Development (RAD) model as the Software Development Life Cycle (SDLC) approach in the design, development, and implementation of the proposed Integrated Student Organization Management System for the National Aviation Academy of the Philippines – Villamor. The SDLC provides a structured framework for developing information systems through defined phases, ensuring systematic requirements analysis, solution design, system development, testing, and deployment while maintaining quality and operational efficiency. Compared with other SDLC models such as Waterfall, which follows a more linear and rigid process, RAD offers greater flexibility through iterative prototyping and continuous user feedback, making it more suitable for systems with evolving requirements.

The RAD model was chosen due to its focus on quick prototyping, iterative development, and ongoing user participation, which are crucial in meeting the changing and evolving needs of managing student organizations. Considering that the current procedures at NAAP–Villamor are manual and disjointed, regular input from the Office of Student Affairs, organization leaders, and students is essential to ensure that the system accurately mirrors actual workflows. RAD provides adaptability in enhancing system capabilities throughout development, shortens overall development duration, and guarantees prompt identification and resolution of problem

The RAD-based SDLC implemented in this study consisted of four major phases: Requirements Planning, User Design, Construction, and Cutover. Each phase was systematically executed to ensure the successful development, validation, and deployment of the proposed system.

Figure 1 illustrates the Rapid Application Development (RAD) model adopted in this study, demonstrating the iterative and incremental progression of system development phases. The model emphasizes rapid prototyping, active user participation, and continuous stakeholder feedback to refine system requirements and functionalities. Through repeated design, development, and evaluation cycles, the system was progressively improved until it met the intended objectives and user expectations.

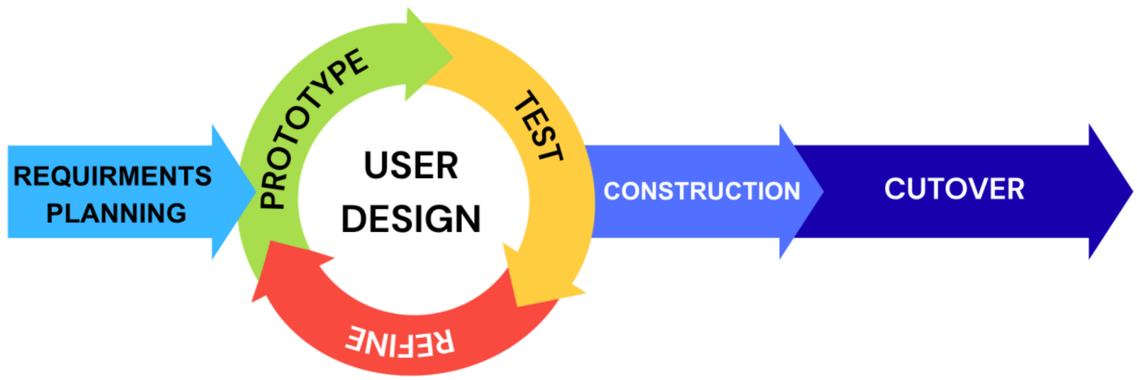

Figure 1. Rapid Application Development

Requirements Planning Phase

During the Requirements Planning phase, the researchers identified and analyzed the problems, objectives, and system requirements of the proposed Integrated Student Organization Management System. An assessment of the existing student organization management procedures was conducted involving the Office of Student Affairs, student organization officers, and students.

This study observed that the current system involved manual submission of documents, physical record keeping, and the use of multiple communication platforms. These practices resulted in delays in approval processes, difficulty in tracking documents, inefficient record management, and limited accessibility of organizational information.

To address these issues, the researchers gathered and defined both functional and technical requirements necessary for system development. The identified system requirements included. Secure login and authentication system, role-based access for students, organization officers, and administrators, digital submission and approval of organizational documents, Organization and membership management module, Notification and communication system, Dashboard and reporting features for monitoring organizational activities.

User Design Phase

In the User Design phase, the researchers designed the system architecture, user interface, and system workflows based on the identified requirements. System prototypes were developed to visualize the layout, features, and overall functionality of the proposed system.

Prototypes of key system components included, login and authentication interface, user dashboards for students, organization officers, and administrators, document submission and approval interface, organization management module, notification and communication features.

Construction Phase

During the Construction phase, the researchers developed the actual web-based Integrated Student Organization Management System based on the approved design and requirements. The system design was translated into a functional application using appropriate programming tools, development frameworks, and database management systems.

The implemented system directly addressed the issues identified in the existing manual and fragmented processes by incorporating key components such as a secure user authentication system, role-based access control, digital document submission and approval, organization and membership management, notification and communication features, and a reporting and analytics dashboard. These components were specifically designed to streamline workflows, improve data accessibility, and enhance administrative efficiency within the Office of Student Affairs and student organizations.

Throughout this phase, continuous system testing was conducted to ensure proper functionality, usability, security, reliability, and efficiency. Identified errors were corrected, and performance improvements were applied to ensure that the system met established quality standards.

Cutover Phase

The Cutover phase marked the final stage of the RAD-based SDLC, where the system was tested, deployed, and implemented for actual use. Final system testing ensured all features functioned properly, while User Acceptance Testing (UAT) was conducted with selected students, organization officers, and administrators to evaluate usability, performance, and requirement compliance. System orientation and demonstrations were also provided to familiarize users with its features and operations. After validation and approval, the Integrated Student Organization Management System was officially deployed.

This phase ensured successful implementation and provided users with a secure, efficient, and accessible platform for managing student organization activities. Through the RAD methodology, the researchers developed a reliable, user-centered system that meets institutional requirements, improves operational efficiency, enhances accessibility, and offers a scalable and centralized solution for student organization management.

Development Methodology Justification

The Rapid Application Development (RAD) model was considered the most appropriate methodology for the proposed Integrated Student Organization Management System because the system requires a development approach that is flexible, user-centered, and capable of producing working results within a limited academic timeframe. Since the system involves multiple functional components such as document management, monitoring, notifications, analytics, and role-based access, RAD is suitable because it supports the rapid development of prototypes that can be reviewed and refined continuously. This allows the researchers to ensure that each feature is practical, responsive to user needs, and aligned with the operational requirements of the Office of Student Affairs, student organization officers, and student members.

RAD is also well suited for the system because it encourages continuous testing, evaluation, and improvement throughout development rather than only at the final stage. This is important for a web-based management system that must be usable, reliable, and efficient across different user roles. Through its iterative nature, RAD allows issues in functionality, usability, and workflow design to be identified early and improved immediately, leading to a more refined and effective final system. For these reasons, the RAD model was the best choice for developing a system that requires adaptability, active user validation, and efficient implementation.

System Development Framework

The System Development Framework outlines the structured methodology used in developing the proposed Integrated Student Organization Management System for the National Aviation Academy of the Philippines – Villamor. The framework is based on the Rapid Application Development (RAD) model under the

Software Development Life Cycle (SDLC), guiding the system through requirements analysis, design, development, testing, and deployment. It ensures a systematic, efficient, and user-centered development process aligned with institutional requirements.

Analysis of the Existing System

The Analysis of the Existing System provides a comprehensive examination of the institution's current student organization management processes, focusing on how records, documents, announcements, activities, and organization-related transactions are handled within the Office of Student Affairs. This section examines the existing procedures, tools, and stakeholders involved in managing student organization operations and administrative coordination. It identifies current limitations and operational challenges such as delays in document submission and approval, fragmented communication, manual record-keeping, limited accessibility to organization-related information and services, and difficulties in monitoring and tracking organizational activities. Through problem analysis, the root causes of these inefficiencies are identified, followed by a requirements gap analysis that compares the present system's capabilities with the functional and non-functional requirements of an effective, secure, and centralized web-based management system. The section also considers possible alternative solutions, evaluates their feasibility in terms of technical, operational, economic, and schedule aspects, and discusses proposed technology alternatives that may improve automation, accessibility, reliability, scalability, and overall system performance, thereby establishing the foundation for the development of an improved and integrated student organization management system.

Current Evaluation Procedures

The management processes of student organizations at the National Aviation Academy of the Philippines – Villamor are presently marked by manual, decentralized, and disjointed methods overseen by the Office of Student Affairs (OSA). The majority of administrative duties depend significantly on paper documents and in-person collaboration. This conventional method limits efficiency and hinders overall processes.

Student organization officers must provide financial reports, accomplishment reports, event proposals, and other relevant documents in physical copy. OSA staff manually examines these submissions to confirm adherence to institutional policies. Consequently, approvals frequently necessitate several in-person visits before final authorization is achieved.

Interaction between student groups and the OSA occurs via distinct and fragmented communication channels. Announcements are spread unevenly, resulting in disparities in information distribution. This disjointed communication framework hinders coordination effectiveness among participants.

Documents are kept either in tangible filing systems or in separate digital files. Due to this configuration, monitoring in real-time and centralized oversight of organizational activities cannot occur. Accessing old documents and confirming records is time-consuming.

Although institutional policies governing student organizations are already established, there is no centralized information system to operationalize them. The absence of a structured digital workflow prevents seamless alignment between governance requirements and administrative execution. Consequently, a gap exists between policy implementation and actual operational practices within the institution.

Limitations and Problem Analysis

The institution's continued reliance on manual and fragmented processes causes delays and inefficiencies in the management of student organizations. Because document submissions, reports, and approval requests are handled through physical routing and face-to-face coordination, processing time becomes longer and more dependent on the availability of specific personnel. As a result, organizational transactions are slowed down, scheduled activities may be delayed, and the overall efficiency of the Office of Student Affairs in handling multiple requests is reduced.

The use of manual tracking methods also creates difficulties in monitoring the progress of submissions and approvals. Since records are often updated separately or followed up through informal communication, it becomes difficult to determine the exact status of documents and identify where delays occur. Consequently, accountability and transparency are weakened because stakeholders may not clearly see who is responsible, what actions are pending, or whether transactions have already been completed.

The absence of a centralized digital archive further affects the consistency and reliability of record management. Because organization-related files are stored through disconnected or paper-based methods, retrieving past reports, verifying historical records, and maintaining complete documentation become more difficult. This leads to possible record loss, duplication, incomplete files, and inaccurate reporting, which in turn reduces the dependability of institutional records and weakens administrative continuity.

The lack of real-time monitoring tools also limits the Office of Student Affairs in effectively supervising multiple student organizations. Because there is no unified platform for tracking activities, submissions, and compliance requirements, monitoring is often delayed and dependent on manual updates. As a result, administrative oversight becomes more reactive than proactive, making it harder to respond quickly to issues, enforce compliance, and produce timely consolidated reports for evaluation and decision-making.

Students are likewise affected by the absence of a centralized platform for announcements, updates, and access to organization-related services. Because information is distributed through fragmented communication channels, some students may receive updates late, miss important announcements, or have limited access to relevant services and opportunities. This communication gap can reduce student awareness, participation, and engagement, thereby affecting the overall effectiveness of student organization management within the institution.

Requirements Gap Analysis

The existing operational landscape shows a notable gap in policy execution within the institution. Although national policies promoting digital transformation and effective governance have been established, existing processes still predominantly depend on manual approaches. This situation highlights a gap between regulatory systems and real institutional operations. To bridge this gap, the suggested web-based Integrated Student Organization Management System aims to implement these policies by digitizing procedures, standardizing workflows, and facilitating efficient, transparent, and policy-compliant administration of student organizations within the institution.

National directives like CHED Memorandum Order No. 09 promote Smart Campus initiatives and the incorporation of technology in educational institutions. In the same way, Republic Act No. 11032 requires streamlined, digital, and effective public service provision. Furthermore, ARTA Memorandum Circular No. 2020-03A strengthens mechanisms for zero-contact and digital transactions in processes related to government.

The present operation reveals a notable gap in policy implementation within the organization. Despite having national policies that advocate for digital transformation and efficient governance, current processes are mainly manual. This scenario underscores a gap between regulatory systems and real institutional behaviors.

National policies like CHED Memorandum Order No. 09 support Smart Campus programs and the incorporation of technology in educational institutions. In the same way, Republic Act No. 11032 requires streamlined, digital, and effective public service provision. Furthermore, ARTA Memorandum Circular No. 2020-03A strengthens the implementation of zero-contact and digital transaction methods in government procedures.

Alternative Solutions and Feasibility Analysis

When assessing possible system options, retaining the current manual system would require no additional development cost; however, it would also preserve the institution's existing inefficiencies, processing delays, and fragmented coordination practices. Because transactions would continue to depend on paper-based submissions, physical routing, and separate communication channels, the institution would remain limited in terms of accessibility, monitoring, and administrative efficiency. In direct comparison, the proposed web-based management system offers a more effective solution because it digitizes and centralizes these processes, resulting in faster transactions, improved coordination, and better record management.

Another alternative is the use of generic or subscription-based third-party platforms. While these systems may provide ready-made functions, many of them require recurring subscription fees, have limited features in free versions, or are not fully aligned with the governance and operational requirements of higher education institutions in the Philippine setting. Their generalized design often makes them less suitable for institution-specific workflows, approval structures, and compliance needs. In direct comparison, the proposed system is more appropriate because it is designed specifically for NAAP–Villamor, allowing the features, workflows, and data structure to directly reflect the institution’s actual student organization processes and administrative requirements.

Generic systems also present limitations in customization, scalability, policy integration, and control over security mechanisms. Since these platforms are built for broad use, they may not fully support the institution’s required approval hierarchies, reporting procedures, and compliance with national digital governance policies. As a result, relying on external systems may create long-term operational restrictions and recurring financial obligations. Compared with these platforms, the proposed system provides greater flexibility and sustainability because it can be developed according to the institution's exact needs while maintaining direct control over functionality, security, and future enhancements.

Considering these factors, the development of a secure, centralized, and institution-specific integrated management system is identified as the most feasible solution. The proposed web-based platform for NAAP–Villamor is intended to support electronic document submission, structured approval workflows, centralized records management, real-time monitoring, and improved access to organization-related services and information. More importantly, it is better than both the manual system and generic third-party platforms because it combines institutional relevance, policy alignment, operational efficiency, and system control in one solution. Furthermore, the system supports compliance with CHED Memorandum Order No. 09, Republic Act No. 11032, and ARTA Memorandum Circular No. 2020-03A, thereby enhancing institutional efficiency, transparency, and regulatory compliance.

Proposed Technology Alternatives

The suggested technological option entails creating a secure, centralized, and institution-tailored web-based Integrated Management System for student organizations at NAAP–Villamor. This system aims to transform disjointed and manual processes into a cohesive digital framework. By means of system integration, processes within the organization will be made more efficient and uniform.

The system will enable electronic document submission, organized approval processes, centralized digital record keeping, and integrated communication channels. By integrating organization-related transactions into one platform, redundant processes will be removed and the need for physical coordination will be reduced. This integration will allow for real-time observation and improved management of submissions and approvals.

The suggested platform will greatly improve administrative supervision by the Office of Student Affairs. Automated processes and unified dashboards will enable proactive oversight of several organizations at the same time. Consequently, transparency, accountability, and operational oversight will be enhanced.

The system will enhance access to information and services for students and organizational officials. Notifications, progress reports, and necessary paperwork will be accessible through a unified secure platform. This centralized access will enhance communication and boost student involvement.

The suggested system is in accordance with national digital governance structures, such as CHED Memorandum Order No. 09, Republic Act No. 11032, and ARTA Memorandum Circular No. 2020-03A. By facilitating efficient, electronic, and no-contact transactions, the platform will implement governance needs into organized digital workflows. This infrastructure aligned with policy will improve efficiency, compliance, transparency, and overall effectiveness in managing student organizations.

Feasibility Study

The feasibility study assesses the practicality of developing and implementing the proposed Integrated Student Organization Management System at the National Aviation Academy of the Philippines – Villamor. It determines whether the project can be effectively carried out within the institution's available resources, operational environment, financial capacity, and target development period. The study is examined through four dimensions: technical, operational, economic, and schedule feasibility.

Technical feasibility examines whether the required technologies, infrastructure, and technical expertise are available for the development and implementation of the proposed system. The study considers the use of a web-based platform, centralized database, secure authentication, role-based access control, digital workflows, and analytics features that are all achievable using currently available development tools and institutional computing resources. Recent studies on web-based management and student information systems show that these technologies are widely applicable and effective in improving administrative efficiency, data accuracy, and system reliability in academic settings. These findings support the technical viability of the proposed system.

Operational feasibility evaluates whether the proposed system can function effectively within the institution's existing environment and user setting. The system is designed to support the current needs of the Office of Student Affairs, student organization officers, and student members by improving record management, workflow coordination, monitoring, and access to information. Because the proposed platform is intended to align with the actual operational processes of student organization management, it is considered suitable for institutional use. Studies on integrated dashboards, automated notifications, and centralized databases also indicate that such systems improve communication, coordination, and decision-making among stakeholders, which supports the operational feasibility of the study.

Economic feasibility determines whether the expected benefits of the proposed system justify the resources required for its development and implementation. Although the project may require expenditures for development tools, hosting, maintenance, and related technical resources, the system is expected to provide long-term value by reducing inefficiencies, minimizing delays, improving record accuracy, and decreasing reliance on manual processes. Compared with repetitive manual coordination and fragmented documentation, a centralized system offers a more sustainable and efficient approach to student organization management.

Schedule feasibility assesses whether the proposed system can be developed and deployed within the given academic timeframe. The study adopts a structured development approach that supports incremental design, testing, and refinement of system features. Since the system focuses on defined core functions within a limited institutional scope, its development is considered achievable within the proposed schedule. This makes the project realistic in terms of time and implementation planning.

To further support the feasibility of the proposed system, recent project studies and related systems from 2022 to 2026 were also considered, particularly those involving web-based management platforms and student information systems. These studies show that centralized systems with secure authentication, controlled access, automated workflows, and analytics features contribute to improved efficiency, communication, and decision-making in academic institutions. Such findings strengthen the practicality and relevance of implementing a similar system in the context of NAAP–Villamor.

In conclusion, the feasibility study confirms that the proposed Integrated Student Organization Management System for NAAP–Villamor is viable and attainable concerning technical, operational, financial, and scheduling aspects. The presence of appropriate technologies, alignment with institutional procedures, expected long-term cost savings, and a practical development timeline collectively suggest that the system can be successfully developed and incorporated into the institution. The results indicate that the suggested system is a practical, sustainable, and advantageous approach for enhancing student organization management, accessibility, and involvement at NAAP–Villamor.

Technical Feasibility

The technical feasibility of the proposed Integrated Management System Enhancing Student Organization Member Accessibility and Engagement at the National Aviation Academy of the Philippines – Villamor is established through the availability of required technologies, infrastructure, and technical competencies necessary for system development and deployment. The proposed system is designed as an online platform, guaranteeing access via common web browsers without the need for specialized hardware or proprietary software. As the institution currently employs internet access and computer labs for both academic and administrative functions, the current infrastructure is adequate for system implementation. The architecture of the platform is based on a client-server model, where users (students, officers of student organizations, and administrators from the Office of Student Affairs) interact with the system through web browsers, and centralized servers manage data processing, authentication, storage, and workflow automation.

From a development standpoint, the system was implemented using widely adopted web technologies to ensure reliability, maintainability, and scalability. The front-end interface was developed using HTML, CSS, and JavaScript, which provided the structural framework, visual design, and interactive functionalities of the web application. On the server side, PHP was utilized as the primary back-end scripting language to manage core system operations, including form processing, business logic implementation, session handling, and overall application control. The system follows a hybrid monolithic architecture, where all core modules are integrated within a single application while maintaining logical separation to improve organization and maintainability.

For data management, SQL was used within a relational database management system to store, retrieve, and manage essential records such as business documents, user accounts, approval transactions, and system logs. PHP was responsible for securely connecting the application to the database to ensure efficient data handling and integrity. During development, XAMPP served as the local hosting environment, providing Apache, MySQL, and PHP in an integrated setup for testing and refinement. For deployment, the system was migrated to a cloud-based hosting platform to enhance accessibility, scalability, and availability. The system can operate on standard personal computers (PCs) or laptops with modern web browsers and stable internet connectivity, making it accessible to students, organization officers, and administrators without requiring specialized hardware.

The system incorporates secure login mechanisms, role-based access control (RBAC), encrypted data transmission (HTTPS), and centralized database management to protect sensitive student and organizational data. Security design considerations align with ISO/IEC 25010 quality characteristics, particularly in terms of security, reliability, performance efficiency, and maintainability. The development approach follows a structured Software Development Life Cycle (SDLC), specifically the Rapid Application Development (RAD) model, which emphasizes iterative prototyping, continuous testing, and user feedback integration. This ensures that technical issues can be identified and resolved early during development, reducing implementation risks.

In terms of deployment, the platform may be hosted on a local institutional server or a secure cloud-based hosting environment, depending on available institutional resources. Both deployment strategies are technically viable, as the system does not require high computational complexity or advanced hardware specifications. Scalability issues can be managed by optimizing servers and implementing database indexing if user access grows over time. Considering the presence of web technologies like HTML, CSS, JavaScript, PHP, and SQL, along with the existing internet infrastructure of the institution, standard hardware capabilities, and the technical skills of Information Systems students and faculty, the suggested system is technically viable, offering accessible software resources, controllable implementation challenges, and an institutional environment that can support sustained deployment and operation.

Operational Feasibility

The operational feasibility of the proposed Integrated Management System for Improving Student Organization Member Accessibility and Engagement at the National Aviation Academy of the Philippines – Villamor is based on the institution's current administrative framework, user preparedness, and coherence with existing governance regulations and operational methods. Currently, the management of student organizations at NAAP–Villamor is carried out through manual, decentralized, and disjointed processes under the oversight of the Office of Student Affairs (OSA). Submission of documents, approvals, oversight, and communications are conducted through tangible paperwork and distinct communication methods, leading to delays, reduced transparency, and heightened administrative burden. The suggested system does not present completely new procedures; instead, it digitizes and enhances current workflows like financial report submissions, accomplishment reporting, event proposals, monitoring, and approval routing.

The system is designed to support three primary user groups: students, student organization officers, and OSA administrators. Role-based dashboards ensure that each user category interacts only with features relevant to their responsibilities, minimizing complexity and reducing training requirements.

Students gain centralized access to organization information, services, and event participation records, while organization officers manage submissions and monitoring tasks. OSA administrators retain oversight authority through verification and approval functionalities. This alignment between system roles and existing institutional roles enhances user acceptance, strengthens willingness to adopt the system, and improves operational compatibility.

Operational feasibility is further strengthened by the system's compliance with Republic Act No. 11032 and ARTA Memorandum Circular No. 2020-03A, which promote zero-contact transactions and streamlined service delivery. By automating document submission, approval workflows, and monitoring, the system operationalizes these mandates within the student organization governance framework. Additionally, alignment with CHED Memorandum Order No. 09, s. 2020 (Smart Campus Initiative) ensures that the system supports institutional digital transformation goals.

In terms of user readiness, the primary stakeholders, students and administrators, are already accustomed to using web-based platforms for academic and communication purposes. This strong familiarity with digital platforms supports clearer user acceptance of the proposed system. The proposed system's web-based interface, notification features, and centralized dashboards are consistent with commonly used digital systems, reducing resistance to adoption. Training requirements are minimal and can be addressed through orientation sessions, user manuals, and guided demonstrations during implementation. Although limitations exist, such as dependency on stable internet connectivity and phased deployment among selected organizations, these constraints do not prevent system operation. Instead, they represent manageable implementation considerations that can be addressed through institutional IT support and gradual rollout strategies.

From an economic perspective, studies on web-based information systems indicate that transitioning from manual to digital processes significantly reduces long-term operational costs associated with paper-based documentation, physical storage, and repetitive administrative labor [58][59]. Initial development costs, including system design, programming, and deployment, are offset by long-term benefits such as improved process efficiency, reduced processing time, and minimized resource consumption. Additionally, the use of open-source technologies and scalable web-based architectures further lowers implementation and maintenance costs while ensuring system sustainability.

The proposed system is operationally feasible because it aligns with existing institutional workflows, supports established governance structures, complies with national policies on digital service delivery, and matches the digital literacy level of its intended users. The system enhances efficiency and oversight without requiring major organizational restructuring, ensuring practical implementation within NAAP–Villamor. Therefore, the proposed system is not only operationally feasible but also highly acceptable to its intended users, making it a practical and sustainable solution for improving student organization management at NAAP–Villamor.

Economic Feasibility

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is economically feasible because the expected long-term savings and efficiency gains outweigh the costs of development, implementation, and maintenance. The system is designed to improve governance efficiency, accessibility, records consolidation, digital workflows, and administrative monitoring through a centralized web-based platform. By replacing manual and fragmented processes with a more organized digital system, the institution can reduce unnecessary administrative effort, shorten processing time, and improve the overall management of student organization transactions. Its scalability also increases its economic value, since the system can accommodate more users, organizations, and records without requiring major additional costs.

The primary costs of the system involve software development, database setup, testing, deployment, and user orientation. However, these costs are justified because the system is developed using the Rapid Application Development (RAD) model, which promotes efficient use of resources through rapid prototyping, iterative refinement, and continuous user feedback. This approach helps prevent wasted time and unnecessary expenses by ensuring that features are aligned early with actual institutional needs. In addition, the user-friendly design of the system reduces the need for extensive training, which further lowers implementation and adjustment costs.

The existing manual process also creates hidden operational costs in the form of time loss, repeated administrative effort, paper-based handling, and inefficient coordination. Physical submissions, fragmented communication, manual record-keeping, and difficulty in tracking documents require more staff time and increase the likelihood of delays, duplication, and human error. These inefficiencies consume institutional resources that could otherwise be used for more productive administrative functions. By digitizing submission, approval, notification, and monitoring processes, the proposed system reduces repetitive tasks, minimizes paperwork, lessens follow-up burdens, and improves coordination, thereby producing clear efficiency gains in daily operations.

The system also provides both tangible and intangible economic benefits. Tangible benefits include reduced paper consumption, lower printing and storage needs, faster transaction processing, and more efficient handling of records and requests. Intangible benefits include improved transparency, better accessibility, stronger accountability, and more reliable data management. Real-time access to records and updates allows users to respond more quickly and make timely decisions, while analytics and reporting tools support better planning and monitoring. These improvements contribute to more efficient use of institutional time, effort, and resources.

Overall, the proposed system is economically feasible because it offers a practical and sustainable solution that reduces operational costs while improving efficiency and administrative effectiveness. Although the institution must make an initial investment in development and implementation, the system is expected to generate long-term value through cost savings, reduced workload, faster processing, and improved resource management. For these reasons, the proposed system is considered cost-justified and beneficial for NAAP–Villamor.

Schedule Feasibility

The schedule feasibility of the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor was evaluated by assessing the development timeline, system testing schedule, and deployment plan in relation to the operational and academic activities of the institution. The assessment aimed to determine whether the system could be designed, developed, tested, and implemented within a timeframe that supports institutional requirements without disrupting student organization processes and administrative functions. The proposed system is intended to improve accessibility, document handling, monitoring, and student engagement through a centralized web-based platform.

The system development follows the Rapid Application Development (RAD) model, which supports fast and iterative development through continuous prototyping, user feedback, testing, and refinement. The SDLC adopted in the study consists of four major phases, namely Requirements Planning, User Design, Construction, and Cutover. Each phase was systematically carried out to ensure that system requirements were identified properly, prototypes were evaluated by intended users, and the final system was tested and deployed within the academic timeframe. The iterative nature of the RAD model also helps reduce the risk of delays by allowing issues to be identified and addressed at an early stage of development.

The implementation schedule was designed to ensure that the system could be deployed before full institutional use so that students, student organization officers, and Office of Student Affairs administrators would have sufficient time for familiarization, validation, and adjustment. This timeline allows adequate time for system testing, User Acceptance Testing (UAT), system orientation, and final improvements prior to actual implementation. Conducting deployment after the completion of development and validation activities helps ensure that the system functions properly and minimizes disruption to document submissions, approval workflows, notifications, and monitoring processes.

Potential scheduling risks include technical issues during development, limited availability of stakeholders during consultation and prototype evaluation, revisions resulting from user feedback, and possible delays in testing and deployment. These risks are mitigated through the RAD approach by supporting continuous stakeholder involvement, iterative refinement, and flexible adjustment of system components throughout the development cycle. The use of a web-based platform also contributes to schedule feasibility because it simplifies deployment and allows users to access system features without complex installation requirements.

Additionally, user orientation and system demonstrations were incorporated into the implementation timeline to support readiness and improve adoption among intended users. This is important because the system involves multiple user groups with different roles and functions, including students, organization officers, and administrators. Since the proposed platform includes role-based access, digital document submission and approval, notification services, and reporting features, providing sufficient time for orientation and validation supports smoother implementation and more effective transition from the previous manual process.

Based on the alignment of the development phases with institutional requirements, the inclusion of testing and orientation activities, and the flexibility of the RAD-based development framework, the proposed system is considered schedule feasible. The structured yet iterative nature of the development process supports timely completion and implementation of the system without significantly disrupting the student organization management operations of NAAP–Villamor.

Cost Benefit Analysis

Overview

This cost-benefit analysis evaluates the economic feasibility of developing and implementing the Integrated Management System Enhancing Student Organization Members' Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor. The analysis compares the estimated development and operational costs of the proposed system with its expected tangible and intangible benefits over a three-year period, which is a common evaluation horizon for academic system feasibility studies.

Cost Analysis

Development Cost

The development cost covers the planning, design, module development, testing, deployment, and initial system setup of the proposed web-based integrated management system.

Table 1. Development Cost

Cost Component	Description	Estimated Cost (PHP)

Requirements Analysis and Sprint Planning	Stakeholder consultation, process mapping, backlog creation, and planning of system iterations	35,000
System Design and Architecture	User interface design, database design, system structure, and module planning	45,000
Core Module Development	Development of user management, organization management, membership records, announcements, events, and document-related modules	150,000
Dashboard and Reports Development	Creation of monitoring tools, summaries, activity reports, and engagement-related dashboards	30,000
Testing and Iteration	Functional testing, usability testing, debugging, revisions, and refinement of module	40,000
Deployment and Configuration	Hosting setup, deployment, environment configuration, and system initialization	20,000
Total Development Cost		**320,000PHP**

Operational and Maintenance Costs (Annual)

These costs represent the yearly expenses needed to maintain the system and ensure its continuous availability, reliability, and usability.

Table 2. Operational and Maintenance Costs (Annual)

Cost Component	**Annual Cost (PHP)**
Web Hosting and Domain Maintenance	30,000
System Maintenance and Bug Fixes	24,000

Security, Backup, and Monitoring	18,000
User Training and Technical Support	12,000
Total Annual Operational Cost	**84,000 PHP**

Total Cost Over 3 Years

Total Cost = Development Cost + (3 × Annual Operational Cost)

Total Cost = 320,000 + (3 × 84,000)

Total Cost = 572,000 PHP

Benefit Analysis

A. Tangible Benefits (Annual)

The proposed system is expected to generate measurable benefits through reduced manual processing time, improved transaction efficiency, lower paper-based expenses, and better use of administrative effort. These benefits are aligned with the broader view that digitization reduces administrative burden by lowering the time and effort required to complete routine processes.

Table 3. Tangible Benefits (Annual)

Benefit Category	Description	Annual Savings (PHP)
Administrative Time Savings	Less time spent on manual encoding, membership validation, record retrieval, and repetitive checking of student organization data	110,000 PHP
Improved	Faster handling of announcements,	60,000 PHP

Processing Efficiency	event coordination, organization requests, and member-related transactions	
Reduced Paper and Printing Expenses	Lower use of printed forms, notices, attendance sheets, and manual reports through digitized processing	25,000 PHP
Reduced Repetitive Follow-up Work	Fewer duplicate submissions, manual clarifications, and repeated status follow-ups because data is centralized and easier to track	40,000 PHP
Better Utilization of Administrative Resources	Existing personnel can devote more time to coordination, student support, and decision-making instead of routine clerical tasks	35,000 PHP
Total Tangible Benefits (Annual)		**270,000 PHP**

B. Intangible Benefits

Although not directly quantifiable in monetary terms, the proposed system is expected to provide significant institutional and operational advantages. It will improve accessibility to organization-related services and information, allowing users to easily access what they need anytime. The system will also increase student engagement and participation in organizational activities by providing a more interactive and user-friendly platform. Communication among student members, officers, advisers, and administrators will become more efficient and streamlined. In addition, the system will enhance the accuracy, transparency, and accountability of organization records and transactions. It will enable faster access to information, supporting effective monitoring, reporting, and decision-making processes. Users will also experience greater convenience and satisfaction through a centralized web-based platform. Overall, the system will contribute to stronger institutional support for digitalization efforts and the modernization of services. These types of benefits are consistent with research on digital public services and GovTech, which emphasizes improved access, better service delivery, and more efficient use of resources through digital systems.

C. Total Benefits Over 3 Years

The total benefits of the proposed system over a three-year period are derived from the annual tangible benefits that the system is expected to generate. These benefits may include cost savings, improved efficiency, reduced manual workload, and other measurable financial gains brought about by the implementation of the system. Given that the estimated annual tangible benefit is 270,000 PHP, the total benefit over three years is calculated by multiplying this value by three. As a result, the system is projected to generate a total of 810,000 PHP in tangible benefits over the three-year period. This cumulative benefit reflects the long-term value of the system and highlights its capacity to continuously deliver measurable advantages to the organization over time.

Total Benefits = 3 × Annual Tangible Benefits

Total Benefits = 3 × 270,000 = 810,000 PHP

D. Net Benefit Computation

The net benefit represents the overall financial gain that the organization will achieve after accounting for all system-related costs. It is calculated by subtracting the total system cost from the total benefits over the same period. In this case, the total system cost over three years amounts to 572,000 PHP, while the total benefits reach 810,000 PHP. By deducting the total cost from the total benefits (810,000 − 572,000), the resulting net benefit is 238,000 PHP. This positive net benefit indicates that the system not only recovers its costs but also generates additional financial value for the organization. It demonstrates that the investment is worthwhile and contributes to the overall financial sustainability and efficiency of operations.

$$\textbf{Net Benefit} = \textbf{Total Benefits} - \textbf{Total Costs}$$

$$\textbf{Net Benefit} = \textbf{810,000} - \textbf{572,000} = \textbf{238,000 PHP}$$

E. Return on Investment (ROI)

Return on Investment (ROI) is a key financial metric used to evaluate the efficiency and profitability of an investment. It measures how much return is gained relative to the cost of the investment and is expressed as a percentage. The ROI is computed by dividing the net benefit by the total system cost and then multiplying the result by 100. Using the given values, the computation is as follows: (238,000 / 572,000) × 100, which results in an ROI of 41.61%. This means that for every peso invested in the system, the organization can expect a return of approximately 41.61 centavos over the three-year period. A positive ROI of this magnitude indicates that the system is financially beneficial and justifies the investment, as it provides a significant return compared to the costs incurred.

ROI = (Net Benefit / Total Cost) × 100

ROI = (238,000 / 572,000) × 100 = 41.61%

F. Cost-Benefit Summary

Table 4. Cost-Benefit Summary

Metric	Value
Total Development Cost	320,000 PHP
Total 3-Year Operational Cost	252,000 PHP
Total System Cost (3 Years)	572,000 PHP
Total Benefits (3 Years)	810,000 PHP
Net Benefit	238,000 PHP
ROI	41.61%

The overall cost-benefit analysis presents a comprehensive view of the financial implications of implementing the proposed system. The total development cost is estimated at 320,000 PHP, which covers expenses related to system design, development, testing, and initial deployment. In addition, the system incurs operational costs amounting to 252,000 PHP over three years, which may include maintenance, updates, hosting, and technical support. When combined, these costs result in a total system cost of 572,000 PHP for the entire three-year period.

On the other hand, the system is expected to generate total benefits of 810,000 PHP over the same timeframe. This leads to a net benefit of 238,000 PHP, indicating that the system produces more value than it consumes in resources. Furthermore, with a calculated ROI of 41.61%, the analysis clearly demonstrates that the proposed system is not only cost-effective but also a profitable investment.

Overall, the cost-benefit summary highlights the financial feasibility and sustainability of the system. It supports the conclusion that implementing the proposed solution will bring long-term economic advantages, improve operational efficiency, and contribute positively to the organization's strategic goals.

Conclusion

The cost-benefit analysis shows that the proposed Integrated Management System Enhancing Student Organization Members' Accessibility and Engagement is economically feasible within a three-year evaluation period. The estimated total benefits exceed the projected development and maintenance costs, resulting in a positive net benefit and a favorable return on investment.

The tangible gains of the proposed system do not depend on reducing the number of administrative personnel. Instead, the system is expected to reduce manual effort, shorten processing time, lessen repetitive follow-up work, and improve the efficiency of existing administrative processes. This means that the value of the system lies in better productivity, improved service delivery, and more effective use of available resources. That approach is consistent with administrative-simplification and digital-service literature, which treats lower burden and better efficiency as core benefits of digitization.

Beyond measurable financial benefits, the system also supports stronger transparency, accessibility, communication, and student engagement in organization-related activities. These benefits further justify the implementation of the proposed system in the National Aviation Academy of the Philippines – Villamor as a practical and institutionally beneficial solution.

System Requirements Specification

This section presents the System Requirements Specification of the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor. It provides a structured presentation of the system requirements necessary for the development and implementation of the proposed web-based platform. Specifically, this section identifies the relevant stakeholders and their corresponding user roles, defines the functional requirements that describe the essential services and operations the system must perform, and outlines the non-functional requirements that establish the standards of quality needed for effective system operation. Through these specified requirements, the study ensures that the proposed system remains aligned with the institutional objectives of improving accessibility, strengthening administrative efficiency, supporting student organization management, and providing a secure, reliable, and user-centered digital environment.

Stakeholder Identification and User Roles

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor involves several key stakeholder groups whose participation is necessary for the effective implementation and operation of the system. Identifying these stakeholders helps ensure that the system's functions are aligned with institutional workflows, governance requirements, and role-based access controls. Since the proposed platform is intended to centralize student organization management, document handling, monitoring, and service accessibility, clearly defined user roles are essential to maintain security, accountability, and operational efficiency.

Students constitute one of the primary stakeholder groups of the system. As intended end-users, they access organization-related information, receive notifications, monitor activities, and engage with services made available through the web-based platform. The system is designed to improve their accessibility to announcements, organizational records, and participation-related information through a centralized and user-friendly environment. Their role in the system is mainly centered on viewing relevant information, receiving updates, and interacting with services that support student organization engagement within the institution.

Student organization officers serve as another major stakeholder group because they are directly involved in the day-to-day management of organizational activities and records. Through the system, they are expected to manage membership-related information, submit required organizational documents, monitor attendance and activities, and coordinate with the Office of Student Affairs through digital workflows. Their role is more operational in nature, as they act as the main organizational users responsible for maintaining updated records and complying with institutional submission and reporting requirements.

Office of Student Affairs (OSA) administrators function as the primary oversight and approval stakeholders within the proposed system. As the office responsible for supervising student organizations, OSA administrators are expected to review submitted documents, monitor compliance, oversee organizational transactions, and use dashboard and reporting tools for decision-making and administrative control. Their access to the system supports more efficient monitoring, better coordination, and improved institutional oversight compared with the current manual and fragmented process. This role is especially important because the study emphasizes governance efficiency, regulatory compliance, and administrative effectiveness in student organization management.

System administrators also play a critical role in ensuring the stability, security, and continuous operation of the platform. Their responsibilities include managing user accounts, maintaining access permissions, safeguarding stored organizational and student data, and ensuring that the system remains functional, secure, and aligned with institutional requirements. Since the proposed system includes secure authentication, role-based access control, and a centralized database, administrative management of technical operations is necessary to preserve data integrity and maintain the reliability of digital services.

Taken together, the identification of these stakeholder groups strengthens the overall design and usability of the proposed system. Role-based access ensures that students, student organization officers, OSA administrators, and system administrators interact only with functions relevant to their responsibilities. This improves security, supports accountability, and promotes more efficient execution of student organization processes. In the context of the study, clearly defined user roles also contribute to the successful implementation of the RAD-based development approach because continuous stakeholder involvement helps ensure that the final system remains responsive to actual institutional needs.

Functional Requirements

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor shall provide the core functions needed to support centralized, efficient, and secure student organization management. These functional requirements define the major system features that will improve accessibility of services, document handling, communication, monitoring, and administrative control within the institution.

To ensure secure access, the system shall provide a login and authentication feature for all authorized users. Students, student organization officers, and administrators shall log in using valid credentials before accessing the system. The platform shall also implement role-based access control so that each user can only access the features and data relevant to their assigned responsibilities.

In support of organization processes, the system shall include a document submission and approval function that allows student organization officers to submit required files digitally and enables administrators to review, verify, and monitor these submissions. The system shall also provide an organization and membership management feature for maintaining organization records, profiles, and related information in a centralized database.

To improve coordination and service delivery, the system shall include a notification feature that sends updates, reminders, and announcements to relevant users regarding activities, submissions, and other transactions. In addition, the platform shall provide dashboard and reporting features that allow administrators to monitor records, track organizational activities, and support decision-making through organized and accessible information.

The system shall also maintain a centralized database for storing user data, organization records, submitted documents, and transaction histories. This shall support more accurate record retrieval, improved monitoring, and better records consolidation compared to the current manual process.

These functional requirements support the development of a secure, centralized, and user-oriented web-based platform for student organization management at NAAP–Villamor. By integrating authentication, role-based access, digital workflows, notifications, reporting, and centralized record management, the proposed system is expected to improve operational efficiency, transparency, accessibility, and administrative effectiveness.

Non Functional Requirements

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor shall meet essential quality standards to ensure that the system operates effectively and supports the institutional needs of students, student organization officers, and administrators. These non-functional requirements focus on the overall quality of the system in terms of security, usability, reliability, performance efficiency, maintainability, compatibility, portability, and accessibility.

In terms of security, the system shall protect user accounts, organizational records, and submitted documents through secure authentication and role-based access control. Only authorized users shall be allowed to access specific features and data relevant to their responsibilities. This is necessary to maintain confidentiality, accountability, and protection of institutional information.

With respect to usability, the system shall provide a clear, organized, and user-friendly interface that enables users to perform tasks easily and efficiently. Navigation, forms, and system processes shall be designed in a way that supports convenient access to records, submissions, notifications, and reports. This requirement is important to ensure that the platform remains practical and understandable for all intended users.

For reliability and performance efficiency, the system shall maintain stable operation and acceptable response time during regular use. It shall be capable of processing submissions, approvals, and notifications without unnecessary delays. The system shall also preserve data accuracy and support dependable record storage to reduce interruptions and maintain smooth institutional operations.

In relation to maintainability, the platform shall be structured in a way that allows easier correction of errors, updating of features, and future enhancement of services. The use of the Rapid Application Development (RAD) model supports this requirement by encouraging iterative refinement, testing, and continuous improvement throughout development.

As for compatibility, portability, and accessibility, the system shall function properly across standard web browsers and commonly used devices within the institution. It shall remain usable in the institution's web-based environment and support convenient access to services and records for authorized users. These qualities are necessary to ensure that the proposed system can be implemented effectively and sustained over time.

These non-functional requirements support the development of a secure, reliable, efficient, and user-centered platform for student organization management at NAAP–Villamor. By meeting these quality standards, the proposed system can better support accessibility, streamline administrative workflows, and improve the overall delivery of student organization services.

Proposed System Architecture and Design

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is designed as a web-based and centralized platform intended to improve the management of student organization records, services, and administrative processes. Its architecture and design were developed to provide a structured, secure, and organized environment that supports accessibility, operational efficiency, and effective coordination among students, organization officers, and OSA administrators/staff. Through an integrated system framework, the proposed platform establishes the foundation for the presentation of its overall architecture and the major system components and modules that enable its operation.

System Architecture Overview

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor adopts a web-based layered system architecture designed to support centralized data management, secure system access, and efficient delivery of student organization services. The architecture enables multiple stakeholders, including students, student organization officers, and OSA administrators/staff, to interact with the system through a web browser or mobile-responsive user interface, while maintaining coordinated application processing, dashboard-based monitoring and reporting, and centralized database storage.

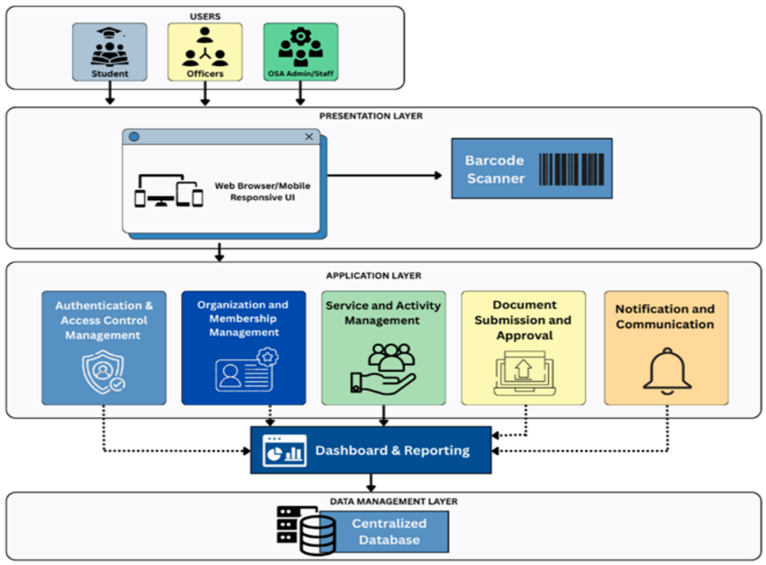

Figure 2. System Architecture of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines - Villamor

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is organized into three major architectural layers, namely the presentation layer, application layer, and data management layer. This layered structure is intended to support secure user interaction, efficient processing of system functions, and centralized storage of organizational data. Through this architecture, the system provides an organized framework for managing student organization services, document transactions, notifications, reporting, and administrative processes within a unified web-based environment.

At the upper level, the presentation layer serves as the user-facing component of the system. It provides access through a web browser or mobile-responsive interface, allowing students, organization officers, and OSA administrators/staff to interact with the platform using standard internet-enabled devices. This layer is designed to support convenient navigation, accessibility, and responsiveness across different screen sizes, making it easier for users to submit documents, manage records, access services, and monitor updates. The inclusion of a barcode scanner in the interface layer also supports faster identification and processing of selected transactions within the system.

At the center of the architecture, the application layer functions as the main processing environment of the system. This layer contains the core modules responsible for authentication and access control management, organization and membership management, service and activity management, document submission and approval, notification and communication, and dashboard and reporting. These modules work together to execute the primary functions of the platform, including secure login, role-based access, handling of organization records, digital submission workflows, delivery of notifications, and generation of reports. The application layer also ensures that each user interacts only with the features relevant to their assigned role, thereby promoting security, accountability, and proper control of system operations.

At the lower level, the data management layer is responsible for maintaining the centralized database of the system. This layer stores user accounts, organization profiles, membership records, submitted documents, communication logs, activity data, and other related transactions necessary for system operation. Centralized storage supports consistency of records, more efficient retrieval of information, and better coordination among system modules. It also strengthens reporting and monitoring functions by ensuring that all processed data are maintained within a single and organized repository.

The layered design supports efficient deployment and continuous operation of the proposed platform. Because the system is web-based, multiple users can access its services at the same time while administrators manage records, approvals, and monitoring tasks from a centralized environment. This setup also makes the system easier to maintain, update, and expand as institutional needs change. The arrangement of modules and database support further contributes to smoother integration of student organization processes and improved digital service delivery.

The proposed system architecture provides a secure, scalable, and organized framework for student organization management at NAAP–Villamor. By combining a responsive user interface, integrated application modules, dashboard-based monitoring, and centralized data storage, the architecture supports improved accessibility, administrative efficiency, and more effective handling of student organization records and services.

System Components and Modules

The proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor is composed of several interconnected modules designed to support centralized student organization management, digital document processing, administrative monitoring, and improved access to organization-related services and information. The modular structure of the system promotes organized system development, easier maintenance, and efficient coordination of institutional processes. Through this design, the system addresses the limitations of manual submissions, fragmented communication, and inconsistent record management by integrating key functions into a unified web-based platform.

The Authentication and Access Control Module serves as the security gateway of the system. This component manages user login, verifies authorized access, and applies role-based permissions for students, student organization officers, and Office of Student Affairs administrators. By limiting access according to user responsibilities, the module helps protect records and transactions while supporting accountability, confidentiality, and secure system use.

The Organization and Membership Management Module functions as one of the core operational parts of the platform. This module maintains organization profiles, membership-related information, participation records, and other essential organizational data within a centralized database. It enables student organization officers and authorized administrators to manage records more efficiently while reducing redundancy and improving accessibility of information.

The Document Submission and Approval Module manages the digital processing of organizational requirements. Through this module, student organization officers can upload reports and other required documents, while Office of Student Affairs administrators can review, verify, approve, and monitor submissions through a structured workflow. This component supports compliance with streamlined and electronic transaction practices by reducing reliance on physical paperwork and improving transparency in document handling.

The Notification and Communication Module is intended to improve coordination among users of the system. This module delivers updates, reminders, announcements, and transaction status notifications related to submissions, approvals, organization activities, and student requests. Its inclusion helps minimize missed communications, reduce repeated follow-ups, and strengthen responsiveness among students, officers, and administrators. Related literature also supports the value of centralized notification channels in improving accountability and institutional coordination.

The Dashboard and Reporting Module provides a consolidated view of organizational information, submissions, and monitored activities. Through this component, administrators can track compliance, review records, observe activity trends, and support decision-making using organized and accessible system data. This module strengthens institutional oversight by transforming stored records into useful reports and monitoring tools that enhance governance efficiency and administrative effectiveness.

The Service and Activity Management Module supports student engagement by allowing users to access organization-related services, monitor participation in activities, and track requests or transactions associated with student organizations. This module reflects the study's goal of improving accessibility and engagement by giving students a more convenient way to interact with organization services, events, and related processes through a single platform.

The Administrative Management Module oversees the overall operation of the platform. Its functions include user account administration, monitoring of system activity, management of institutional records, and maintenance of system settings necessary for continuous operation. This module helps ensure that the system remains organized, secure, and aligned with institutional workflows and operational requirements.

Supporting these modules is the Centralized Database Component, which stores user information, organization records, submitted documents, transaction histories, and activity logs. This component serves as the primary repository of system data and is essential in maintaining data accuracy, record consolidation, and efficient retrieval of organizational information. The use of a centralized database also supports monitoring, reporting, and secure records management throughout the platform.

Taken together, these system components enable the proposed platform to operate as a secure, centralized, and user-oriented management system for NAAP–Villamor. The modular architecture supports maintainability, scalability, and efficient integration of student organization processes while remaining consistent with the study's objectives and the RAD-based SDLC adopted for system development.

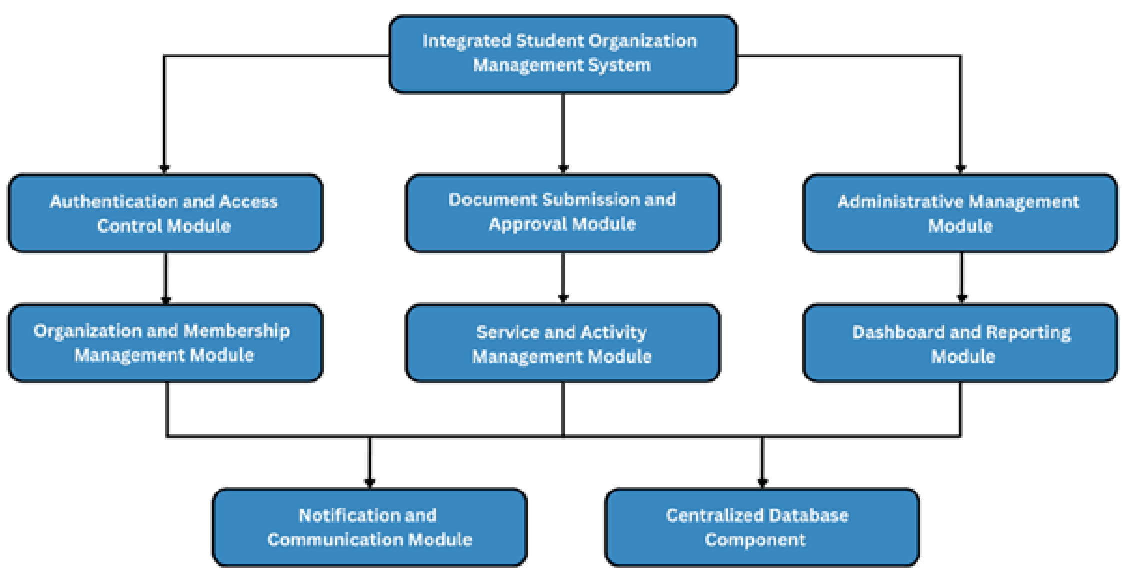

Figure 3. Proposed Modular Architecture of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines - Villamor

Figure 3 presents the modular architecture of the proposed Integrated Student Organization Management System. The Integrated Student Organization Management System serves as the core platform that coordinates the major system functions and supports the centralized management of student organization processes. Connected to the core are the Authentication and Access Control Module, which ensures secure user login and role-based access permissions, the Document Submission and Approval Module, which manages the digital submission, review, and approval of organizational requirements, and the Administrative Management Module, which oversees system administration, user management, and operational control. Supporting these primary modules are the Organization and Membership Management Module, which handles organization profiles, membership records, and related organizational data, the Service and Activity Management Module, which manages student organization services, activities, and participation-related processes, and the Dashboard and Reporting Module, which provides consolidated records, monitoring tools, and reports for administrative decision-making. At the lower level of the architecture are the Notification and Communication Module, which delivers announcements, updates, and transaction-related notifications, and the Centralized Database Component, which serves as the main repository of system data, records, and transaction histories. The modular and layered arrangement of these components supports scalability, maintainability, clarity of process flow, and efficient integration of the proposed system.

Process Modeling and System Workflow

This section presents the process modeling and system workflow of the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National Aviation Academy of the Philippines – Villamor. It provides a structured representation of how the system operates by illustrating its overall system boundary, the flow of data between external stakeholders and internal processes, the interaction of major processing components and data stores, and the sequence of activities performed during system use. Through the context diagram, Data Flow Diagram (Level 0), Data Flow Diagram (Level 1), and procedural workflow, this section explains how the proposed platform manages user access, organization-related transactions, document handling, event and attendance processes, rental services, and administrative monitoring within a centralized and organized web-based environment.

Context Diagram

The context diagram presents the overall system boundary and the interactions between the proposed Integrated Management System Enhancing Student Organization Members Accessibility and Engagement in the National

Aviation Academy of the Philippines – Villamor and its external stakeholders. In the diagram, the system is treated as a single centralized process that exchanges information with the primary users involved in student organization management at NAAP–Villamor, particularly students, student organization officers, Office of Student Affairs (OSA) administrators, and system administrators. This high-level view emphasizes the system's role as a unified web-based platform for improving accessibility, coordination, document handling, monitoring, and administrative efficiency.

Students interact with the system by accessing organization-related information, receiving notifications, monitoring activities, and using organization-related services made available through the platform. Student organization officers interact with the system by managing organization and membership records, submitting required documents, monitoring activities, and coordinating transactions through digital workflows. These interactions reflect the study's goal of improving student access to services, records, requests, and updates through a centralized platform.

OSA administrators engage with the system by reviewing submitted documents, monitoring compliance, overseeing organization-related transactions, and using dashboard and reporting tools for decision-making and administrative control. System administrators, on the other hand, interact with the platform by managing user accounts, maintaining access permissions, monitoring system activity, and ensuring the security and continuous operation of the platform. These roles support the study's emphasis on stronger oversight, secure access, and centralized management of student organization processes.

The context diagram does not present the internal modules or detailed processing logic of the system. Instead, it focuses only on the external entities and the major data exchanges that occur between them and the proposed platform, such as submissions, approvals, notifications, records, monitoring information, and administrative controls. This representation is important because it helps define the scope of the system, identify stakeholder participation, and clarify the flow of information needed for the design and development of the proposed web-based student organization management system.

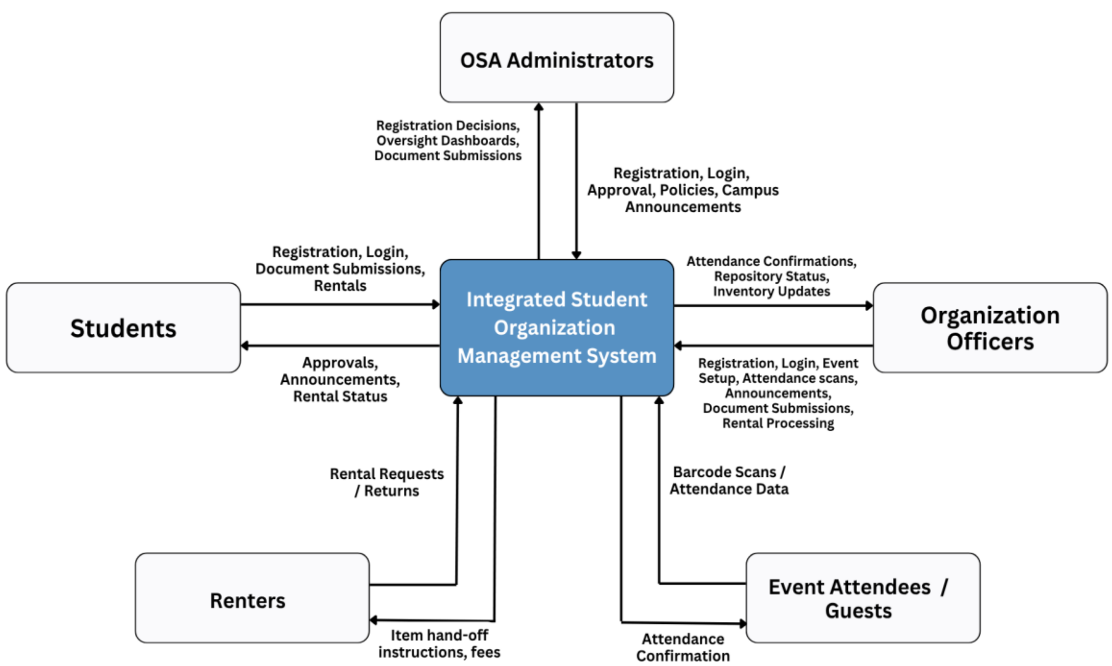

Figure 4. Context Diagram of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines – Villamor

Figure 4 illustrates the context diagram of the proposed Integrated Student Organization Management System. The diagram presents the system as a centralized process that interacts with external stakeholders, including students, student organization officers, and OSA administrators. It highlights the exchange of organization-related information, document submissions, activity and service records, notifications, monitoring data, and administrative control information between the users and the system, thereby defining the system boundary and the scope of stakeholder interaction.

Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram presents the major data exchanges of the proposed Integrated Student Organization Management System and shows how the system interacts with its primary external users. In the diagram, the system serves as the central platform that receives, processes, and returns information related to registration, login, document submission, approvals, announcements, attendance, and rental transactions. This representation provides a high-level view of how information moves between the system and the stakeholder groups involved in student organization activities and related services.

From the perspective of students, the system accepts data such as registration details, login credentials, document submissions, and rental-related requests. In return, the system provides outputs including approvals, announcements, and rental status updates. This flow indicates that students use the platform not only for account access but also for receiving official updates and monitoring the status of their transactions within the organization management process.

In relation to organization officers, the system receives inputs associated with registration, login, event setup, attendance scans, announcements, document submissions, and rental processing. After processing these transactions, the system returns information such as attendance confirmations, repository status, and inventory updates. This shows that organization officers have a more operational role in the system because they handle a wider range of organization-related activities and records.

For OSA administrators, the diagram shows that the system supports administrative and oversight functions. The system receives inputs involving registration, login, approvals, policies, and campus announcements, while it provides outputs such as registration decisions, oversight dashboards, and document submission information. This data flow reflects the role of OSA administrators in supervising organization processes, reviewing transactions, and monitoring institutional compliance through the platform.

As shown in the diagram, event attendees or guests interact with the system mainly through barcode scans or attendance data, while the system returns attendance confirmation when applicable. This exchange supports event monitoring and participant validation. In the same way, renters interact with the platform by submitting rental requests and returns, while the system provides item hand-off instructions and fee-related information. These flows demonstrate that the system also functions as a support platform for event participation and rental-related transactions beyond standard organization records management.

The Level 0 Data Flow Diagram illustrates the system as a centralized digital platform that connects multiple stakeholder groups through structured data exchange. It defines the broad scope of system interaction without presenting lower-level internal processing details, thereby helping clarify how the proposed Integrated Student Organization Management System manages organization-related information, approvals, attendance, announcements, and rental transactions in a unified environment.

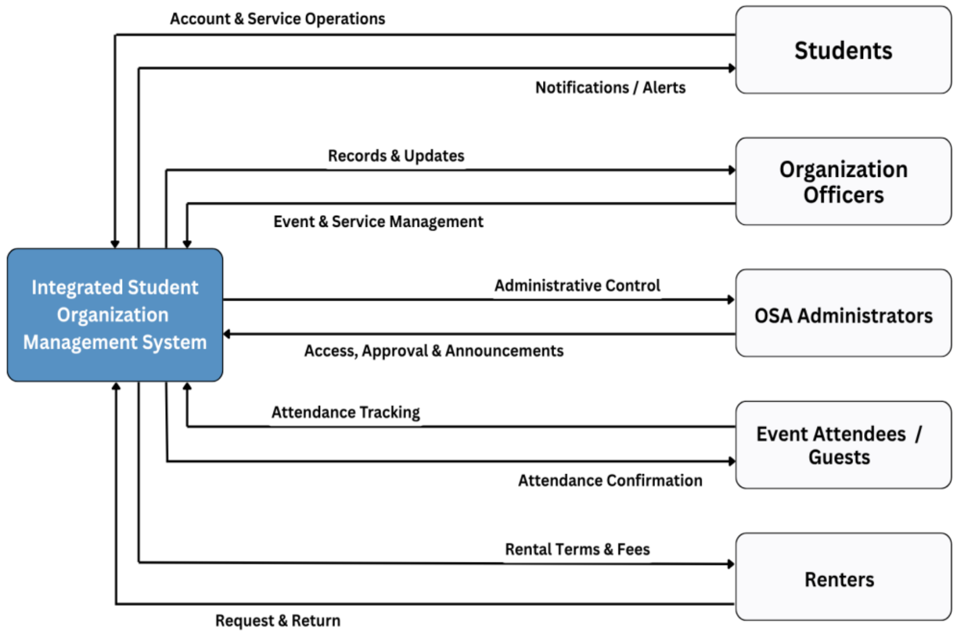

Figure 5. Data Flow Diagram (Level 0) of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines - Villamor

Figure 5 illustrates the Level 0 Data Flow Diagram of the proposed Integrated Student Organization Management System. The diagram presents the system as a centralized process that interacts with external stakeholders, including students, organization officers, OSA administrators, event attendees or guests, and renters. It highlights the exchange of registration and login data, document submissions, approvals, announcements, attendance information, rental transactions, and administrative control data between the users and the system, thereby defining the scope of interaction and the major flow of information within the proposed platform.

Data Flow Diagram Level 1

The Level 1 Data Flow Diagram presents a more detailed view of the internal processes of the proposed Integrated Student Organization Management System by breaking down the major functions shown in the Level 0 DFD into more specific processing components. Through this diagram, the flow of data between the system processes, data stores, and external entities becomes clearer. It shows how the platform handles user account transactions, event and attendance activities, document-related processes, and rental operations while maintaining coordination with students, organization officers, OSA administrators, event attendees or guests, and renters.

At the first level of processing, the Manage Users and Accounts (1.0) process handles registration and login transactions submitted through the system. This process is responsible for receiving account-related inputs, validating user access, and maintaining account information through the User Accounts DB (D1). It supports the system's need for controlled access by ensuring that users are properly registered and authenticated before they can proceed with other transactions. Because students and other stakeholders begin their system interaction through registration and login, this process serves as one of the primary entry points of the platform.

The second process, Event Setup and Attendance Processing (2.0), focuses on event-related activities managed within the system. It receives data associated with event setup and attendance operations, particularly from organization officers and attendance-related transactions. The process stores and retrieves relevant records using the Events Repository (D2). Through this component, the system is able to generate outputs such as attendance confirmations, repository status, and inventory-related updates, which support the operational responsibilities of organization officers and the monitoring functions required by the institution.

In a similar manner, the Document Repository Processing (3.0) component manages document submission transactions and related record handling. This process accepts document inputs, supports the review and routing of submitted files, and stores the processed records in the Document File Store (D3). It also contributes to the production of outputs needed by OSA administrators, such as document submission information, registration decisions, and oversight-related data. This part of the diagram highlights the role of the system in organizing document workflows and maintaining a centralized repository of institutional files associated with student organization activities.

Another important process shown in the diagram is Manage Rentals and Inventory (4.0), which deals with rental transactions and inventory-related records. The system receives rental requests, returns, and processing inputs, then manages these transactions using the Inventory and Rentals DB (D4). As a result, the system can provide outputs such as item hand-off instructions and fee-related information to renters. This process indicates that the proposed platform is not limited to record management and document handling alone, but also supports operational services connected to organizational resources and rental activities.

The Level 1 diagram also illustrates how external entities interact with the internal processes of the system. Students primarily exchange data related to registration, login, document submissions, rentals, approvals, announcements, and rental status. Organization officers are associated with event setup, attendance scans, announcements, document submissions, and rental processing, while also receiving attendance confirmations, repository status, and inventory updates. OSA administrators interact with the system through approval-related and oversight-related transactions, while event attendees or guests participate through barcode scans or attendance data and receive attendance confirmation when applicable. Renters communicate with the system through rental requests and returns, then receive instructions and fee-related outputs after processing.

The Level 1 Data Flow Diagram provides a more specific representation of how the proposed Integrated Student Organization Management System processes information internally. It demonstrates how the platform distributes transactions across key functional processes and corresponding data stores while maintaining consistent interaction with external users. By presenting the detailed movement of account data, event information, documents, attendance records, and rental transactions, the diagram strengthens the understanding of how the proposed system operates as an integrated and centralized digital platform for student organization management.

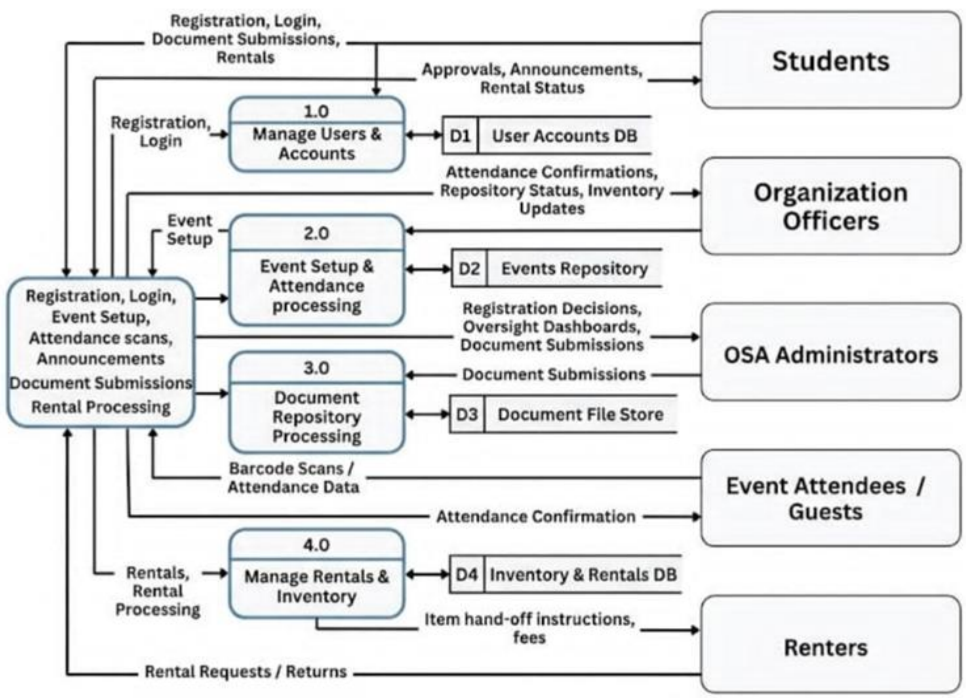

Figure 6. Data Flow Diagram (Level 1) of the Integrated Management System Enhancing Student Organization Members and Accessibility and Engagement in National Aviation Academy of the Philippines – Villamor

Figure 6 presents the Level 1 Data Flow Diagram of the proposed Integrated Student Organization Management System. The diagram illustrates the detailed internal processes of user and account management, event setup and attendance processing, document repository processing, and rental and inventory management. It also shows the exchange of data among external stakeholders, system sub-processes, and the corresponding data stores, including the User

Accounts DB, Events Repository, Document File Store, and Inventory and Rentals DB, thereby providing a more detailed representation of the internal operations of the proposed system.

Procedural Workflow

The procedural workflow of the proposed Integrated Student Organization Management System describes the sequence of activities performed by the users and the corresponding system responses during the management of student organization transactions. It explains how information is entered, processed, stored, and returned through the platform to support registration, login, event setup, attendance processing, document submission, announcements, approvals, rental transactions, and administrative monitoring. Through this workflow, the system provides a centralized and organized process for handling the major operations related to student organization management.

At the beginning of the procedure, the user accesses the system through the web-based platform and submits the required registration or login information. The system then processes these inputs through the Manage Users and Accounts function, where the data are validated and checked against the User Accounts Database. If the submitted information is valid, the user is granted access to the system according to the assigned role. This step ensures that students, organization officers, OSA administrators, and other authorized users are directed only to the functions relevant to their responsibilities.

After successful access, the user proceeds to the appropriate transaction based on the assigned role and current activity. For students, the workflow may involve document submission, viewing announcements, checking approval status, or initiating rental-related transactions. For organization officers, the procedure may include event setup, attendance scanning, posting announcements, submitting organizational documents, and processing rentals. For OSA administrators, the workflow involves reviewing registrations, monitoring submitted files, issuing approvals, managing policies, and viewing oversight dashboards. These activities serve as the main operational inputs received by the system.

Once a transaction is initiated, the system routes the data to the corresponding internal process. Registration and login requests are handled by Manage Users and Accounts. Event-related and attendance-related activities are processed through Event Setup and Attendance Processing, which stores and retrieves relevant information from the Events Repository. Document-related transactions are directed to Document Repository Processing, where submitted files are recorded and maintained in the Document File Store. Rental requests, returns, and related inventory activities are managed through Manage Rentals and Inventory, which uses the Inventory and Rentals Database for storage and retrieval. In this stage, the system validates, updates, and organizes the information according to the nature of the transaction.

As processing continues, the system generates the necessary outputs for the user concerned. Students receive outputs such as approvals, announcements, and rental status updates. Organization officers obtain attendance confirmations, repository status, and inventory updates. OSA administrators receive registration decisions, oversight dashboards, and document-related information needed for administrative control. Event attendees or guests may receive attendance confirmation when barcode scans or attendance data are successfully processed. Renters, on the other hand, receive item hand-off instructions and fee-related information after their rental transactions are completed. These outputs ensure that each stakeholder receives timely and relevant information from the system.

At the same time, the processed data are stored in the appropriate data repositories to support record consistency, retrieval, and future monitoring. Account-related data are kept in the User Accounts DB, event and attendance information are stored in the Events Repository, submitted files are maintained in the Document File Store, and rental and inventory records are managed in the Inventory and Rentals DB. This organized data storage allows the system to maintain accurate records, support administrative monitoring, and provide updated information whenever needed.

The procedural workflow demonstrates how the proposed system coordinates user access, transaction handling, internal processing, data storage, and output generation within a single platform. By integrating account management, event and attendance processing, document repository handling, and rental and inventory control, the system supports a more efficient, accessible, and structured approach to student organization management at NAAP–Villamor.

Data Modeling and Database Design

This section presents the data modeling and database design of the Integrated Student Organization Management System. It explains how institutional data is structured, organized, and interconnected to support governance, event monitoring, documentation control, and rental services within a centralized web-based platform. The section begins with the conceptual data model, which defines the major entities and their relationships at a high level, followed by the Entity Relationship Diagram (ERD) that details the logical database structure, including keys, attributes, and relationship cardinalities. Together, these models provide a systematic foundation for implementing a reliable, scalable, and integrity-driven database architecture.

Conceptual Data Model

This conceptual data model presents the high-level data structure of the Integrated Student Organization Management System by identifying the major entities involved in user management, academic structure, organization governance, event monitoring, documentation control, and rental services. The model defines how institutional data is organized and how different components interact within a centralized web-based platform, without yet specifying technical database implementation details.

At the core of the model is the Users entity, which represents all system stakeholders including students and authorized personnel. Users contain personal, academic, and account-related attributes that support identification, authentication, and activity tracking. The Student Number entity is linked to Users to ensure that each student has a unique institutional identifier. This strengthens traceability and ensures that all transactions, registrations, and records are properly associated with verified individuals.

The academic structure is represented through Institutes and Academic Program. Institutes offer academic programs, and students are enrolled in specific programs. This relationship organizes users according to their academic affiliation and supports administrative monitoring across institutional units. By structuring data this way, the system enables filtering and reporting based on institute or program classification.

Student organization governance is centered on the Organization entity. Each organization contains identifying and descriptive information and serves as the primary unit for events, announcements, rentals, and documentation. The Org

Role entity defines roles within organizations, specifying leadership structures and access privileges. These roles are assigned through Organization Member, which links Users to organizations and defines their responsibilities. The Program Org Mapping entity connects organizations to specific academic programs, formally documenting institutional alignment and organizational scope.

Event management is handled through the Event entity, which stores details of activities hosted by organizations. Students request participation through Pending Registration, and approved participation is recorded in the Attendance Record. Attendance records maintain verified logs of event participation and serve as a basis for engagement monitoring and reporting. This structure replaces manual attendance systems with a centralized digital tracking mechanism.

Communication within the system is facilitated by the Announcement entity. Organizations define, author, and issue announcements that are linked to members and visible to users. This ensures structured dissemination of official information and maintains documented communication history within the platform.

The documentation workflow is represented by Document Submission, Document Annotation, and Approved Document Report. Organizations submit documents that undergo review and annotation before approval. Document annotations allow comments, revisions, and feedback to be recorded systematically. Once approved, documents are stored and summarized into approved document reports, which include relevant organizational and academic references. This structured approval process enhances compliance monitoring, transparency, and accountability in organization-related documentation.

A major operational component of the system is the rental service module. The Rental entity records rental transactions initiated by organizations or authorized users. Rentals are associated with Inventory Category, which groups items based on classification. Each category contains multiple Inventory Item records representing individual rentable resources. Rental transactions include items through the Rental Item entity, which specifies quantity, rates, and item references. This design enables accurate tracking of borrowed resources, monitoring of availability, and maintenance of inventory control.

The relationships among these entities establish a centralized and integrated data environment. Users are linked to academic structures, organizations, events, rentals, and documentation processes. Organizations manage events, issue announcements, submit documents, and oversee rental operations. Documentation passes through structured review and approval stages, while rental transactions are directly connected to inventory records. Attendance, membership, and program mappings provide additional layers of accountability and traceability.

Overall, the conceptual data model supports governance efficiency, regulatory compliance, centralized information access, and administrative effectiveness. By organizing system components into clearly defined entities and structured relationships, the model provides a coherent foundation for the logical and physical design stages of the Integrated Student Organization Management System.

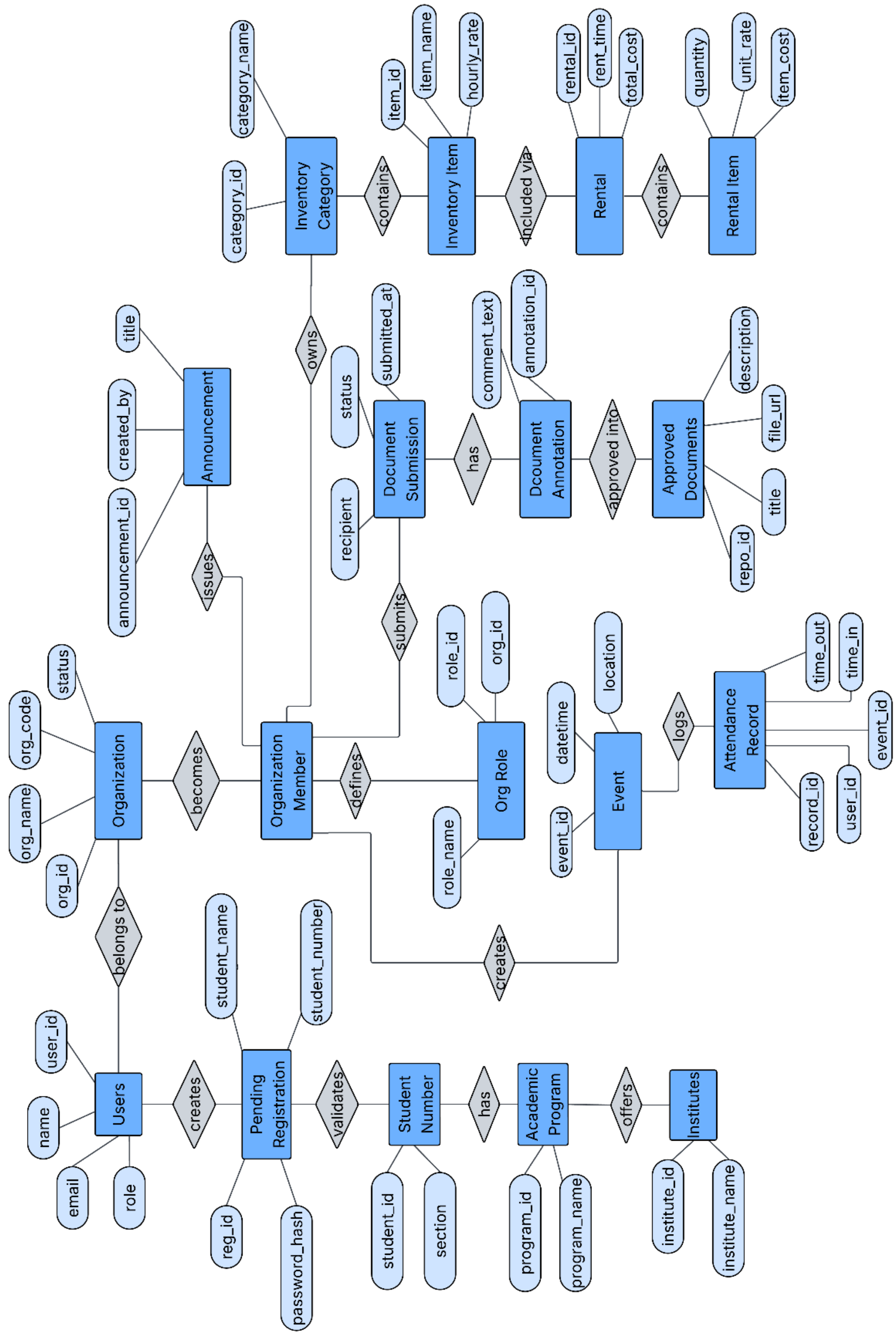

Figure 7. Conceptual Data Model

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure of the Database for the Integrated Student Organization Management System. It defines the entities, their attributes, primary keys, foreign keys, and relationship cardinalities to ensure data integrity, consistency, and controlled processing of organization governance, event management, documentation workflows, and rental services. The ERD establishes how data is interconnected within a centralized institutional platform.

The Users entity serves as the core entity of the system. It stores account and identification details such as user_id, name, email, and account_type. Users participate in multiple system processes, including creating events and announcements, submitting and reviewing documents, processing rentals, and recording attendance. The relationship between Users and other entities is primarily one-to-many, as a single user can create multiple events, announcements, rentals, or document records.

The academic structure is defined through Institutes, Academic Programs, and Student Numbers. An institute has a one-to-many relationship with academic programs, meaning one institute can offer multiple programs. Student Numbers are linked to both institutes and users, ensuring that each student record is uniquely identifiable and institutionally classified. This structure enforces referential integrity by ensuring that student records are associated with valid academic units.

The Organizations entity represents recognized student organizations. Each organization can have multiple Organization Members, establishing a one-to-many relationship between organizations and memberships. Organization Members link users to organizations and specify roles and join dates. Organizational roles are further structured through the Org Roles entity, which defines role names within each organization. This separation ensures role standardization and prevents duplication of role definitions.

Program alignment is maintained through Program Org Mappings, which connect organizations to specific academic programs. This relationship supports institutional oversight by formally associating organizations with academic classifications.

Event management is structured around the Events entity. Each event is created by a user and belongs to one organization, forming a many-to-one relationship from events to organizations and users. Participation is tracked through Attendance Records, which reference both event_id and user_id as foreign keys. This creates a many-to-one relationship between attendance records and events, as well as between attendance records and users. The design ensures that attendance entries cannot exist without valid event and user references.

Student participation requests are handled through Pending Registrations, which store registration data linked to users and organizations. This entity supports controlled approval workflows before final participation is recorded.

The rental module is defined by Rentals, Rental Items, Inventory Categories, and Inventory Items. A rental record is associated with one organization and may involve both a renter and a processor (users), forming multiple foreign key relationships to the Users entity. Each rental can contain multiple rental items, establishing a one-to-many relationship between rentals and rental items. Rental Items reference Inventory Items, which are grouped under Inventory Categories. Inventory Categories belong to organizations, forming a one-to-many relationship between organizations and categories, and between categories and inventory items. This layered structure ensures accurate tracking of item ownership, classification, and usage while preventing orphan inventory records.

Documentation control is managed through Document Submissions, Document Annotations, and Documents Approved. A document submission is linked to the submitting and reviewing users and to a specific organization. One document submission can have multiple document annotations, forming a one-to-many relationship that supports structured review and feedback processes. Once approved, records are stored in Documents Approved, which references the original submission and the approving user. This ensures traceability from submission to final approval and enforces accountability through foreign key constraints.

Announcements are represented by the Announcements entity. Each announcement is created by a user and associated with an organization, establishing many-to-one relationships that ensure announcements cannot exist without valid organizational and user references.

Throughout the ERD, primary keys uniquely identify each entity, while foreign keys enforce referential integrity across related entities. Cardinalities such as one-to-many and many-to-many (resolved through associative entities like Organization Member and Rental Items) ensure structured data relationships and prevent duplication or inconsistency.

By defining structured entity relationships, primary and foreign key constraints, and appropriate cardinalities, the ERD supports reliable data storage, efficient reporting, controlled documentation workflows, accurate inventory tracking, and centralized governance monitoring. The diagram provides the logical foundation for implementing a consistent and scalable database design for the Integrated Student Organization Management System.

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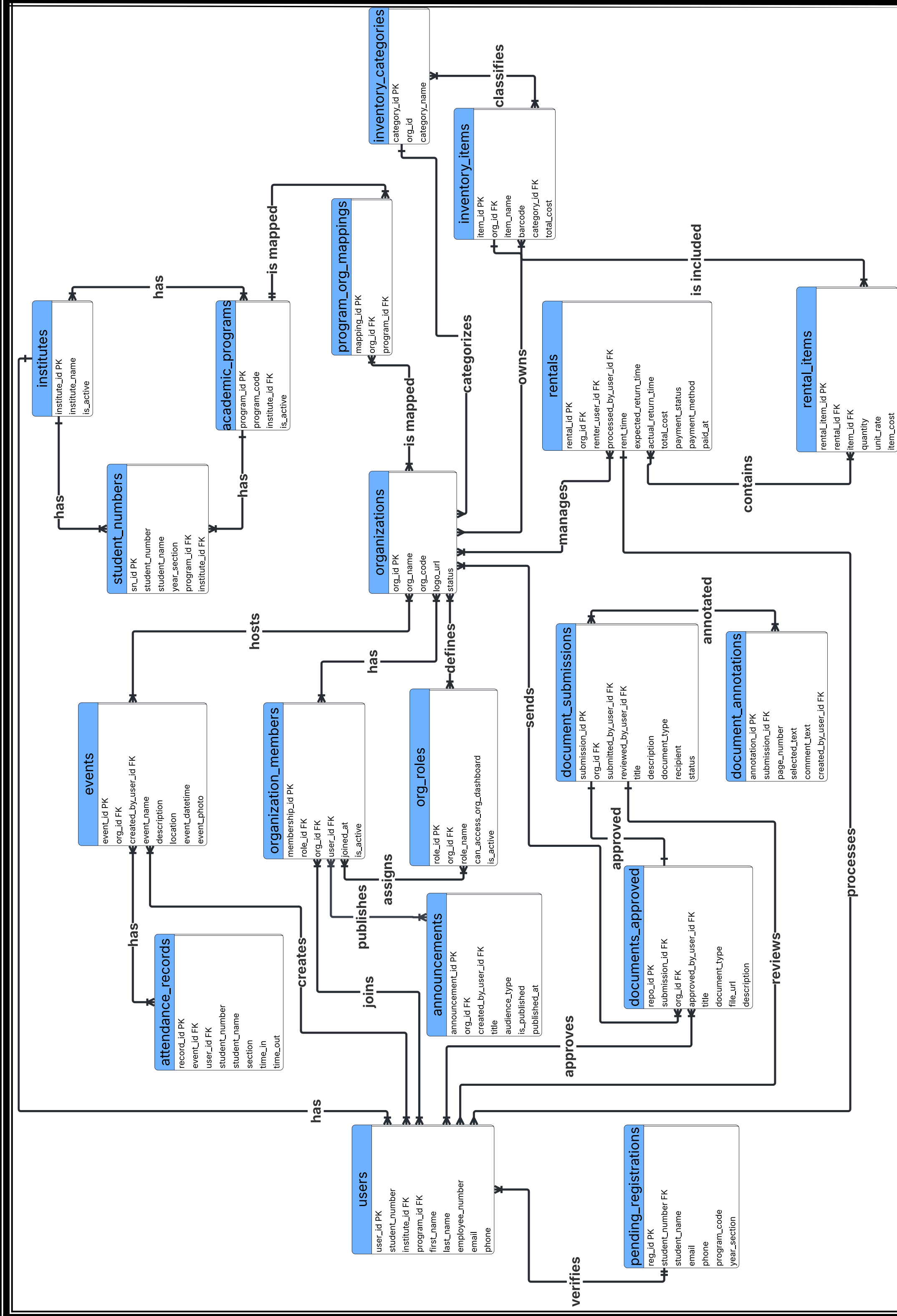

Figure 8. Entity Relationship Diagram

Logical Database Design

The logical database design defines the structured organization of data within the Integrated Student Organization Management and Rental System. The logical model translates the conceptual framework into relational database tables by specifying attributes, primary keys, foreign keys, and relational constraints. This structure ensures data consistency, integrity, and efficient processing of student organization governance, document workflows, event management, attendance monitoring, and rental services within the institution.

The database is composed of multiple interrelated tables grouped into core functional modules: user and academic structure management, organization governance, document processing, event and attendance tracking, and inventory rental management. Each table stores specific data entities while maintaining structured relationships through primary and foreign key constraints to enforce referential integrity.

The Users table serves as the foundational entity for authentication and role management. It stores user identification details, institutional affiliation, academic program reference, contact information, encrypted credentials, account type, and activity status. Each user is uniquely identified by a primary key. Foreign key relationships link users to Institutes and Academic Programs, ensuring that all accounts are institutionally classified and program-specific.

Academic structure is defined through the Institutes, Academic Programs, and Program Org Mappings tables. Institutes represent academic divisions, while Academic Programs are linked to specific institutes. The Program Org Mappings table connects organizations to academic programs, enabling structured alignment between student organizations and their corresponding academic departments. This modular design ensures scalability and prevents redundancy in academic data classification.

Student identity management is supported through the Student Numbers table, which links student records to programs and institutes. This structure ensures controlled student validation and prevents duplication of student information across organizational and attendance modules.

Organization governance is structured through the Organizations, Organization Members, and Org Roles tables. The Organizations table stores organization profile information, including name, logo, status, and timestamps. The Organization Members table establishes many-to-many relationships between users and organizations, allowing users to hold membership records. The Org Roles table defines role classifications within each organization, enabling role-based access and administrative permissions. Foreign key constraints ensure that membership and role assignments correspond to valid users and organizations.

The Events table manages organization-related activities, storing event details such as name, description, location, schedule, publication status, and associated organization references. Event creation is linked to authorized users through foreign keys, ensuring accountability. The Attendance Records table records participant time-in and time-out logs, linking attendance to events and users. This relational structure supports accurate attendance tracking and activity validation.

Document workflow management is handled through the Document Submissions, Documents Approved, and Document Annotations tables. The Document Submissions table stores uploaded documents, including metadata such as document type, title, semester, academic year, and submission timestamps. Review and approval processes are tracked using foreign key references to reviewing users. The Documents Approved table stores finalized records of approved submissions, ensuring a separate structured archive. The Document Annotations table supports review transparency by storing feedback comments and structured annotation data linked to specific submissions. This separation of concerns enhances auditability and preserves document lifecycle integrity.

System announcements are managed through the Announcements table, which stores organization-based announcements, publication status, and target audience classification. Foreign key relationships ensure that announcements are associated with valid organizations and creators.

The rental and inventory management module consists of Inventory Categories, Inventory Items, Rentals, and Rental Items tables. Inventory Categories classify rentable assets. Inventory Items store item details, including barcode, cost, category classification, and payment tracking fields. The Rentals table records transaction-level rental data, including rental duration, return timestamps, total cost, payment method, and transaction status. The Rental Items table supports a many-to-many relationship between rentals and inventory items, allowing multiple items per transaction. This normalized design ensures accurate tracking of item availability, payment status, and overtime calculations.

Pending registration processes are managed through the Pending Registrations table, which temporarily stores applicant data awaiting administrative review. This structure separates unverified accounts from the primary Users table, maintaining system integrity and preventing unauthorized access.

The logical database design applies normalization principles up to the Third Normal Form (3NF). Each table stores logically independent data attributes, eliminating redundancy and ensuring that non-key attributes depend solely on the primary key. Many-to-many relationships are resolved through associative tables such as Organization Members and Rental Items, preventing data duplication and maintaining structural clarity.

Primary and foreign key constraints enforce referential integrity across all modules. These constraints prevent orphan records, invalid references, and inconsistent data entries. Cascading relationships and structured indexing further support efficient query processing and transactional reliability.

Overall, the logical database design provides a scalable, normalized, and relationally consistent foundation for managing student organization governance, digital document workflows, attendance tracking, announcements, and rental services. By structuring data into logically defined tables with enforced relational constraints, the system ensures accuracy, maintainability, performance efficiency, and institutional data governance support.

Figure 9. Logical Database Design

System Security and Data Privacy Design

The proposed Integrated Management System for the National Aviation Academy of the Philippines – Villamor (NAAP–Villamor) is designed to centralize student organization management while ensuring strong security controls, data protection, and regulatory compliance. Since the platform handles student information, organizational records, rental transactions, approval workflows, monitoring dashboards, and administrative data, security and privacy mechanisms are embedded directly into the system architecture. The design emphasizes controlled access, encryption, privacy safeguards, and auditability to ensure governance efficiency, accountability, and institutional data protection.

Access Control and Authentication Mechanisms

The system adopts a role-based and identity-centered access control model to ensure that users can only access functions and information relevant to their institutional responsibilities. Users are categorized into defined roles, including students, student organization officers, Office of Student Affairs (OSA) administrators or approvers, and system administrators. Each role is assigned specific permissions based on institutional duties. Students are limited to viewing organization-related information, browsing available services, and performing rental transactions for items offered by student organizations. Student organization officers are authorized to manage their organization's records, manage services offered (such as rentable equipment, materials, or other organization-owned resources), monitor rental transactions, and upload required organizational reports. OSA administrators are responsible for reviewing and approving organizational reports, monitoring compliance dashboards, and overseeing institutional governance. System administrators are limited to managing technical configurations and maintaining the platform and are not permitted to alter official approval decisions. This separation of duties prevents conflicts of interest, reduces the risk of unauthorized actions, and strengthens accountability within digital workflows.

Authentication mechanisms are implemented to verify user identity before granting access. All users must log in using unique institutional credentials. Passwords are securely stored using non-reversible hashing algorithms such as Argon2id or bcrypt to protect against credential compromise. Multi-Factor Authentication (MFA) is required for privileged roles, including OSA administrators and system administrators, to provide an additional layer of protection against unauthorized access. Secure session management practices are enforced, including HTTP-only and secure cookies, automatic session timeouts after periods of inactivity, and re-authentication for high-risk actions such as approving reports or modifying user roles. These controls reduce the likelihood of unauthorized access and ensure traceable electronic transactions. Account lifecycle management is also implemented, ensuring that user roles are updated or revoked immediately when organizational positions change, thereby maintaining accurate authorization boundaries.

Data Encryption and Protection Measures

Because the system stores confidential student information, organizational records, rental transaction data, and official reports, a defense-in-depth encryption strategy is applied to protect both data confidentiality and integrity. All communication between users and the web application is secured using HTTPS with TLS 1.3 encryption to prevent interception or tampering during transmission. Legacy protocols are disabled to minimize vulnerabilities associated with outdated encryption standards.

Data stored within the system is protected through encryption at rest. The relational database is encrypted, and uploaded organizational documents are stored in encrypted object storage repositories. Backup files are likewise encrypted using separate encryption keys to ensure that stored data remains protected even in recovery scenarios. Encryption keys are managed through secure key management practices, including restricted administrative access, periodic key rotation, and separation of encryption keys from application code. Database credentials, API keys, and other sensitive configuration data are stored securely and are not embedded directly in source code. These measures significantly reduce the risk of large-scale data exposure in the event of unauthorized access.

To preserve the integrity of governance records, the system prevents unauthorized modification of finalized reports and approved organizational records. Once an approval decision has been recorded, the workflow state cannot be altered without creating a new version entry. Hashing or digital verification techniques may be applied to finalized records to detect tampering. These integrity controls ensure that approval histories, compliance records, and rental monitoring data remain reliable and verifiable for administrative oversight.

Data Privacy and Ethical Considerations

While centralization improves accessibility and monitoring, it also increases privacy risks by consolidating student-related information and rental transaction data into a single digital environment. To address this, the system follows privacy-by-design principles. Data minimization is strictly enforced, meaning only information necessary for student organization management, rental processing, service monitoring, and administrative oversight is collected. Each data element has a clearly defined purpose to prevent unnecessary data accumulation.

Role-scoped visibility ensures that personal and organizational data is accessible only to authorized users. Students cannot access confidential organizational reports or internal records of other organizations. Officers can only manage and view records related to their own organization and its offered services. OSA administrators may access identifiable information strictly for governance, compliance, and monitoring purposes. Analytics dashboards prioritize aggregated data to minimize exposure of personally identifiable information and reduce privacy risks.

Retention policies are established to prevent indefinite storage of personal data. Rental transaction records and organizational documents are archived or securely deleted after defined institutional retention periods. Procedures are also implemented to address data access, correction, and complaint requests in accordance with applicable privacy regulations. In the event of a security incident, breach response protocols include detection, containment, documentation, and notification procedures. These measures ensure that digital operations enhance institutional efficiency while respecting individual privacy rights.

Audit Logging and Activity Monitoring

To strengthen transparency, traceability, and compliance, the system incorporates comprehensive audit logging and activity monitoring mechanisms. The platform automatically records login attempts, role assignments and modifications, service creation or updates, rental transactions, workflow status transitions, approval actions, and administrative configuration changes. Each log entry includes the user identifier, timestamp, action performed, and reference to the affected record.

Audit logs are stored separately from primary application data and are protected from modification or deletion by standard users. Access to logs is restricted to authorized administrators for compliance verification and incident investigation. Suspicious activities, such as repeated failed login attempts or unauthorized privilege changes, trigger system alerts for review. Logs are encrypted and retained according to institutional retention policies, and secure backups are maintained to ensure availability during recovery scenarios.

Through structured audit logging, encrypted storage, secure authentication, and controlled authorization, the system establishes a secure and accountable digital environment. These security and privacy mechanisms ensure that the Integrated Management System not only improves administrative efficiency and service accessibility but also protects student, rental, and institutional data in accordance with modern security standards and regulatory requirements.

Implementation Environment and Technology Stack

The implementation environment and technology stack form the technical foundation of the integrated student organization management system. In higher education digital platforms, the selection of appropriate software technologies, hardware configuration, architectural design, and deployment infrastructure significantly influences system performance, security, scalability, and maintainability. For a centralized web-based system that supports rental services, role-based access control, monitoring dashboards, and administrative governance, the chosen environment ensures secure transaction processing, efficient data management, and accessible service delivery across multiple user devices.

Software Requirements

The Integrated Student Organization Management System for NAAP–Villamor operates as a browser-based web application on a PHP/MySQL backend. The software stack is selected to support secure authentication, role-based access, server-rendered pages, and responsive user interfaces.

A web server environment (Apache with PHP) hosts the application, manages client–server communication, and executes business logic for authentication, rentals, approvals, and dashboards. PHP handles server-side rendering for core pages.

A relational database management system (MySQL) stores user accounts, role assignments, rental transactions, organizational records, approval states, and audit entries. The database enforces structured schemas, primary/foreign keys, and uses prepared statements for secure data access.

Client-side technologies include HTML5 for structure, CSS for responsive and consistent styling, and JavaScript for interactive behavior and asynchronous API calls (fetch/AJAX). Modern web browsers (desktop and mobile) are required to render pages, handle forms, and execute client-side scripts securely over HTTPS.

Security controls are supported across the stack: HTTPS/TLS for encrypted transport in production, PHP session management with secure cookies, password hashing (bcrypt/Argon2id), and server-side authorization checks on all API endpoints. Configuration separates environments (development vs. production) so that production instances run with HTTPS, secure cookies, and restricted secrets.

Supporting tools include package/dependency management for client libraries where needed, and backup/monitoring utilities for database and application uptime. This software environment provides a stable, maintainable foundation for centralized organization management, rentals, approvals, and governance workflows.

Hardware Requirements

The deployment of the Integrated Student Organization Management System for NAAP–Villamor requires hardware capable of supporting secure operations, concurrent user access, and reliable processing of organizational records, rentals, and approvals. The hardware baseline prioritizes availability, performance, and data reliability for production use.

The system is hosted on server-class hardware (or an equivalent cloud instance) that runs the Apache/PHP/MySQL stack. The server provides sufficient

CPU resources to handle simultaneous sessions from students, organization officers, and OSA administrators, along with real-time dashboard updates and file uploads. Adequate system memory is provisioned to support PHP application execution, database caching, and responsive page/API delivery during peak activity.

Persistent storage uses enterprise-grade or cloud-backed volumes to retain user accounts, role assignments, rental transactions, organizational documents, and approval records. Storage is sized for current usage with headroom for growth and supports fast read/write performance. Regular backups and, where available, redundant storage configurations are employed to protect against data loss and to enable recovery.

Network infrastructure provides stable, low-latency connectivity between client devices and the server environment. Bandwidth is sufficient to accommodate concurrent logins, data submissions, and document uploads, with encrypted transport (HTTPS) for all production traffic. Network controls restrict administrative interfaces to authorized users and protect the application from unauthorized access.

Client devices require only standard hardware capable of running a modern web browser. Desktops, laptops, tablets, and smartphones can access the system without specialized equipment, benefiting from responsive layouts for varying screen sizes.

By provisioning capable server resources, reliable storage with backups, secure and performant networking, and browser-capable client devices, the hardware environment supports the system's scalability, availability, and secure handling of organizational and student data.

Development Tools and Programming Technologies

The system adopts a monolithic hybrid architecture that combines server-rendered pages and unified backend structure. In this configuration, core modules—such as authentication, rental management, service administration, workflow monitoring, and audit logging—operate within a single application codebase while maintaining logical modular separation. Server-rendered pages manage most user interface rendering directly through PHP. This hybrid approach supports efficiency, simplified deployment, and cohesive integration between rental services and administrative functions.

Front-end development uses HTML, CSS, and JavaScript to create structured, interactive, and responsive interfaces. JavaScript enhances user experience by enabling dynamic form validation, real-time updates. The backend, powered by PHP, enforces authentication, manages session control, processes rental transactions, validates inputs, and communicates securely with the MySQL database. The relational database structure ensures normalized storage of user data, rental records, service listings, and monitoring logs. This technology stack balances simplicity, maintainability, scalability, and security, making it appropriate for a centralized academic management platform.

Deployment Environment

During development, the system operates within a local server environment using XAMPP, which integrates Apache, MySQL, and PHP into a unified hosting package. Apache serves as the web server, PHP executes backend logic, and MySQL manages database operations. This environment enables controlled testing, debugging, and validation of authentication processes, rental workflows, and administrative monitoring features prior to production release.

For operational deployment, the system utilizes a cloud-based hosting environment that supports scalability, remote accessibility, and centralized data management. The deployment configuration maintains Apache, PHP, and MySQL components within a secure cloud infrastructure. Cloud hosting supports HTTPS configuration, encrypted communication, automated backups, and consistent uptime reliability. It also enables authorized users to access the system remotely across desktops, laptops, tablets, and mobile devices. This deployment approach reduces infrastructure maintenance overhead while accommodating growth in rental transactions, organizational data, and user accounts.

Overall, the implementation environment and technology stack consist of HTML, CSS, and JavaScript for the client interface; PHP and MySQL for backend processing and database management; a monolithic hybrid architecture combining server-rendered pages and using XAMPP (Apache + MySQL + PHP); and cloud-based deployment for production use. This integrated technological configuration supports centralized rental service management, role-based access control, secure transaction processing, multi-device accessibility, and sustainable digital governance within a higher education institution.

Chapter 4

RESEARCH METHODOLOGY RESULT AND DISCUSSION

This chapter presents the methodological framework, system implementation results, and analytical evaluation of the proposed Integrated Student Organization Management System developed for NAAP–Villamor. It describes the research design, data collection procedures, system development process, and evaluation methods employed to ensure a systematic and objective assessment of the proposed web-based platform. The chapter further explains the instruments used to measure system performance, functionality, usability, and overall effectiveness in supporting student organization governance and administrative processes.

Following the presentation of the methodology, this chapter discusses the findings derived from system testing, user evaluation, and software quality assessment based on established standards such as ISO/IEC 25010. The results are analyzed in relation to the study's objectives to determine the system's capability to improve information accessibility, streamline digital workflows, enhance monitoring of organizational activities, and strengthen administrative efficiency. Through a structured presentation of empirical evidence and analytical discussion, this chapter establishes the performance, reliability, and practical value of the developed system, serving as a foundation for the conclusions and recommendations of the study.

Research Design

This study employed a Research and Development (R&D) design supported by a mixed-method approach to systematically design, develop, and evaluate the proposed Integrated Student Organization Management System at the National Aviation Academy of the Philippines–Villamor (NAAP–Villamor). The R&D design was adopted as the primary research framework because the study focuses on the creation of a functional system that addresses identified institutional problems while ensuring that it meets established quality, usability, and performance standards. This approach aligns with the nature of the study, which involves both system development and evaluation.

The study began with a descriptive analysis of the existing student organization management processes, particularly the manual, decentralized, and fragmented administrative procedures currently practiced within the Office of Student Affairs. This phase allowed the researchers to identify key operational limitations, including delays in document processing, fragmented communication channels, limited monitoring capability, and the absence of centralized records. These identified issues served as the basis for the design and development of the proposed web-based integrated management system intended to streamline workflows, enhance accessibility, and improve administrative oversight.

The system development component of the study was guided by the Rapid Application Development (RAD) model under the Software Development Life

Cycle (SDLC). This development approach enabled iterative design, prototyping, testing, and refinement of the system through continuous user feedback. The process included requirements planning, user design, system construction, and deployment, ensuring that the developed system is aligned with institutional needs and functional requirements.

For system evaluation, the study adopted a mixed-method approach combining quantitative and qualitative data collection techniques. Quantitative data were gathered using structured survey questionnaires and User Acceptance Testing (UAT) forms administered to students, student organization officers, and administrators. The ISO/IEC 25010 software quality model was used as the primary framework for evaluating system characteristics such as functionality, usability, reliability, performance efficiency, and security. Descriptive statistical tools, including frequency distribution, percentage, mean, and weighted mean, were utilized to analyze numerical data and determine the overall system quality and effectiveness.

In addition to quantitative evaluation, qualitative data were collected through user feedback and recommendations to capture user experiences, insights, and suggestions for system improvement. This qualitative component allowed the researchers to gain a deeper understanding of user satisfaction, usability concerns, and potential enhancements, ensuring a more comprehensive evaluation of the system.

The study also included a development output component, which involved the creation of a user manual designed to guide users in effectively navigating and utilizing the system. This supports system usability, user adoption, and operational sustainability.

Overall, the research design integrates system development and evaluation by combining the R&D framework, RAD-based system development methodology, and mixed-method evaluation approach. This ensures that the study not only produces a functional and institutionally relevant system but also provides empirical evidence on its quality, usability, efficiency, and user acceptability within NAAP–Villamor.

Data Gathering Technique

Data for this study will be collected using a combination of system-generated records, structured survey instruments, and qualitative user feedback to evaluate the proposed Integrated Student Organization Management System at NAAP–Villamor. The data gathering will be conducted during the actual pilot use of the web-based platform by its intended stakeholders—students, student organization officers, and Office of Student Affairs (OSA) administrators—since the system is designed to centralize organization information, services, events, and records while supporting monitoring and coordination through role-based access.

For platform effectiveness and workflow performance, quantitative operational data will be gathered from system logs and database records, including timestamps and counts of document submissions and approvals (e.g., financial and accomplishment reports), tracking and archiving status, processing turnaround time, and frequency of administrative actions. These records will support the study objective of evaluating whether digital workflows improve submission, tracking, and archiving in alignment with RA 11032 and ARTA Zero-Contact requirements, and whether monitoring features strengthen oversight and document management efficiency.

For the "services" component of the system, the data gathering will focus on student rentals (service requests) rather than student document submissions. Transaction data will be collected from service/rental request forms submitted by students through the system, including the type of item/service rented, request dates, approvals/confirmations (if applicable), completion status, and request history. These system-generated service/rental records will be used to evaluate whether the centralized platform improves accessibility, visibility, and management of student organization services and requests, consistent with the scope that allows students to avail, manage, and track organization services with real-time updates.

To evaluate system quality, a structured survey instrument anchored on the ISO/IEC 25010 software quality model will be administered after participants use the platform. The survey will use a Likert-scale format to measure the system's functional suitability, usability, reliability, performance efficiency, security, and overall user satisfaction, aligned with the study objectives on quality evaluation.

To capture user experience issues and improvement recommendations, qualitative feedback will be collected through open-ended questions embedded in the post-use evaluation (or a short feedback form). Responses will be analyzed thematically (e.g., common usability difficulties, clarity of workflows, perceived transparency, and suggestions for improving rental/service requesting and monitoring). This supports the objective of assessing acceptability through User Acceptance Testing (UAT) among the intended user groups.

All data gathering procedures will be handled with confidentiality and privacy safeguards appropriate for student and institutional records. Access to raw data will be limited to authorized roles, and reported results will be aggregated so that individual identities are not exposed, consistent with the system's role-based approach and the need to protect information handled within centralized platforms.

Statistical Treatment of Data

The data gathered in this study will be analyzed using appropriate descriptive, reliability, and comparative statistical techniques to ensure an objective interpretation of findings and to determine whether the proposed Integrated Student Organization Management System effectively addresses the identified problems in student organization management at the National Aviation Academy of the Philippines–Villamor. The statistical treatment is aligned with the objectives of the study, which focus on evaluating the system in terms of accessibility and records consolidation, effectiveness of digital workflows, contribution of role-based controls to administrative oversight, software quality based on ISO/IEC 25010, and operational effectiveness and user acceptability through User Acceptance Testing (UAT).

For quantitative data, the study will use frequency, percentage, mean, weighted mean, and standard deviation. These tools are appropriate for summarizing survey responses and system evaluation results gathered from students, student organization officers, and Office of Student Affairs administrators. In addition, Cronbach's Alpha will be used to test the internal consistency of the evaluation instrument, while percentage improvement may be applied, where applicable, to compare selected operational indicators of the proposed system with the existing manual or fragmented process. If open-ended comments or suggestions are collected during User Acceptance Testing, these will be organized through thematic analysis to support the interpretation of quantitative findings.

1. Frequency (f): Frequency refers to the number of times a particular response occurs in the dataset.

$$f = \text{number of times a response occurs}$$

In this study, frequency will be used to determine how many respondents selected each response option in the evaluation questionnaire. This will help identify the response distribution for indicators related to accessibility, workflow effectiveness, administrative monitoring, software quality, and user acceptability.

2. Percentage (%): Percentage refers to the proportion of respondents who selected a particular response relative to the total number of respondents.

$$P = \frac{f}{N} \times 100$$

Where:

P = percentage
f = frequency of responses
N = total number of respondents

Percentage will be used to present the respondent distribution and summarize the proportion of responses per item or category. This is useful in describing how the students, student organization officers, and Office of Student Affairs administrators evaluated the proposed system.

3. Mean (Arithmetic Mean): The mean is the arithmetic average of a set of numerical values.

$$\bar{X} = \frac{\sum X}{N}$$

Where:
$\bar{X}$ = mean
$\sum X$ = sum of all values
N = number of observations

The mean will be used to compute average values for selected system performance indicators and operational metrics, such as report processing time, response turnaround, or other measurable outputs generated during testing and evaluation. This will help describe the overall operational effectiveness of the system.

4. Weighted Mean: The weighted mean is a measure of central tendency used for Likert-scale responses, where each response is assigned a corresponding weight.

$$WM = \frac{\sum fx}{N}$$

Where:
WM = weighted mean
$\sum fx$ = sum of the product of frequency and weight
N = total number of responses

The weighted mean will serve as the primary statistical tool for interpreting responses in the system evaluation questionnaire. It will be used to assess:

- The extent to which the centralized platform improves accessibility and consolidation of student organization information, services, events, and records;
- The effectiveness of digital workflows in the submission, tracking, and archiving of reports;
- The contribution of role-based access controls and monitoring features to administrative oversight and document management efficiency;
- The quality of the proposed system based on ISO/IEC 25010 in terms of functionality, usability, reliability, efficiency, security, and user satisfaction; and
- The operational effectiveness and user acceptability of the system through User Acceptance Testing (UAT).

Because the main evaluation data of the study will come from scaled responses of intended users, the weighted mean is the most suitable tool for determining the overall level of agreement and acceptability of the proposed system.

The following scale will be used in interpreting the weighted mean:

Scale	Range	Verbal Interpretation
5	4.21 – 5.00	Strongly Agree / Highly Acceptable
4	3.41 – 4.20	Agree / Acceptable
3	2.61 – 3.40	Neutral / Moderately Acceptable
2	1.81 – 2.60	Disagree / Less Acceptable
1	1.00 – 1.80	Strongly Disagree / Not Acceptable

5. Standard Deviation (SD): Standard deviation measures the extent of dispersion or variability of responses around the mean.

$$SD = \sqrt{\frac{\sum(X - \bar{X})^2}{N}}$$

Where:
SD = standard deviation
X = individual value
$\bar{X}$ = mean
N = number of observations

Standard deviation will be used to determine the consistency of the respondents' evaluations. A low standard deviation indicates that the responses are closely clustered around the mean, while a high standard deviation indicates greater variation in perceptions. This is useful in examining whether the three respondent groups show similar or differing views regarding the quality and acceptability of the system.

6. Percentage Improvement: Percentage improvement measures the relative amount of change between two values and expresses it as a percentage.

$$PI = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100$$

Where:
PI = percentage improvement

Where applicable, this tool will be used to compare selected operational indicators of the proposed system with the existing manual or fragmented process. It may be applied to metrics such as processing time, document tracking efficiency, report submission turnaround, and administrative workload. This will help determine whether the proposed system provides a measurable improvement in operational performance.

7. Cronbach's Alpha: Cronbach's Alpha is a statistical measure used to determine the internal consistency or reliability of a survey instrument.

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum \sigma_i^2}{\sigma_t^2}\right)$$

Where:
α = Cronbach's Alpha
k = number of items
$\sum \sigma_i^2$ = sum of the variances of individual items
σ_t^2 = variance of the total score

Cronbach's Alpha will be used to test the reliability of the evaluation questionnaire, particularly the items related to ISO/IEC 25010 quality characteristics and user acceptability. This will ensure that the instrument consistently measures the intended constructs of the study.

8. Thematic Analysis for Open-Ended Responses: If qualitative feedback, suggestions, or comments are gathered from respondents during User Acceptance Testing, these will be analyzed through thematic analysis. Responses will be organized, grouped into recurring themes, and interpreted to identify common issues, recommendations, and user experiences related to the proposed system.

This qualitative treatment will support the quantitative results by providing deeper insight into how users perceive the system's accessibility, usability, efficiency, and overall acceptability.

Population and Sampling Technique

The study will use a purposive sampling technique, selecting participants who directly interact with the proposed system during its implementation at NAAP–Villamor. Purposive sampling is appropriate because the study evaluates an applied, institution-specific platform intended for actual use by defined stakeholders, and meaningful feedback requires respondents who have hands-on experience with the system's functions for services/rentals, event and information access, and administrative monitoring.

The target population includes (1) students who use the system to access organization information, view updates, join activities, and submit rental/service requests; (2) student organization officers who manage organization content, services/rentals, events, and required organizational documents; and (3) OSA administrators who perform oversight functions such as account verification and approval/validation of submitted organizational documents. These groups align with the system's intended users and with the study objectives focusing on accessibility and consolidation, workflow effectiveness, role-based monitoring, system quality, and UAT-based acceptability.

Inclusion criteria will require that respondents (1) are officially part of NAAP–Villamor and belong to one of the identified stakeholder groups, (2) have valid system access credentials based on their role, and (3) have completed at least one relevant system interaction during the pilot period (e.g., students submitting at least one rental/service request or using the service tracking features; officers encoding/managing services and organizational records; OSA staff reviewing/approving or monitoring submissions). Exclusion criteria will cover users with no actual system usage during the evaluation period or incomplete responses that prevent reliable assessment of ISO/IEC 25010 survey results and UAT feedback.

Through purposive sampling, the study ensures that the collected data reflects real system usage and contextually valid evaluation of how the integrated platform supports NAAP–Villamor's student organization governance, service/rental accessibility, monitoring, and administrative effectiveness.

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